

A binomial count of meromorphic travelling waves

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Abstract

We study meromorphic travelling waves of $u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0$, a constant-coefficient family containing KdV–Burgers, Kuramoto–Sivashinsky and Kawahara. After integration and normalization, the profiles satisfy $P(D)v + v^2/2 = 0$, where $D = d/dz$ and P is monic of degree $p \geq 2$ with complex coefficients.

For evolution equations of even order (p odd), and for purely dispersive equations (only odd-order derivatives), we give a complete algebraic description of all nonconstant globally meromorphic travelling waves: they are rational, rational in an exponential, or elliptic, with one pole per period.

One-pole waves of these three kinds are exceptional: for $p \geq 3$ a generic equation of the family has none. We count the exceptions as pairs (P, v) , over the whole family: there are $\binom{2p-1}{p-1}$ with v rational in an exponential of given rate, and $2\binom{2p-2}{p-2}$ with v elliptic on a given generic lattice. We also count the solitary pulses of the purely dispersive subfamily and the fronts of the purely dissipative one (only even-order derivatives in the linear part).

When $2p-1$ is prime, these statements are proved and all pairs are distinct. We conjecture them in every order.

Keywords. Meromorphic travelling waves; dispersive and dissipative equations; elliptic functions; polynomial convolution; algebraic multiplicity.

1 Introduction

Exact travelling waves of nonlinear evolution equations are usually found one equation at a time, and usually only for special values of the coefficients. The Kuramoto–Sivashinsky equation with dispersion, $u_t + uu_x + u_{xx} + \sigma u_{xxx} + u_{xxxx} = 0$, is a classical example. It has nonconstant travelling waves that are polynomials in a hyperbolic tangent for exactly four values of the dispersion coefficient, up to its sign: $\sigma = 0, 4, 12/\sqrt{47}$ and $16/\sqrt{73}$. Kuramoto and Tsuzuki found the wave of the dispersionless equation [61], Kudryashov the others [31–34], and Eremenko proved that, together with Kudryashov’s elliptic solution and a single rational one, they exhaust the nonconstant meromorphic travelling waves [22]. The classical tables list six cases at these four values [6, 42]: two at $\sigma = 0$, two at $\sigma = \pm 4$, and one at each of the other two values. The four values already appear together in Kudryashov [32, Eq. (39)]. The tables do not distinguish a wave from its mirror image, which solves the equation with the opposite sign of σ . If the two are counted separately, the two cases with $\sigma = 0$ are their own mirror images and the other four are not, so the list has $2 + 2 \cdot 4 = 10$ members. This paper shows that ten is $\binom{5}{2}$: the classical list is the case $p = 3$ of a binomial enumeration, proved for infinitely many p and conjectured for all.

We consider the evolution equations

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0,$$

which contain the KdV–Burgers ($p = 2$), Kuramoto–Sivashinsky ($p = 3$), Kawahara and Benney–Lin ($p = 4$) and Nikolaevskiy ($p = 5$) equations; Table 1 lists the named members with $p \leq 6$. Kudryashov studied this family as a whole and found exact waves of its KdV–Burgers, Kuramoto–Sivashinsky and Kawahara members [32]. A travelling wave $u = w(x - ct)$ satisfies, after one integration, $\sum_j b_j w^{(j)} - cw + w^2/2 + K = 0$. Shifting w by a constant solution and normalizing the leading coefficient gives

$$P(D)v + \frac{1}{2}v^2 = 0, \quad P(\rho) = \rho^p + \sum_{j=0}^{p-1} a_j \rho^j, \quad p \geq 2, \quad (1.1)$$

with $D = d/dz$ and complex coefficients. The coefficients $a_1, \dots, a_{p-1}$ are those of the evolution equation, up to a rescaling of x , and the speed, the integration constant and the background are absorbed into a_0 (Section 2). Throughout, p is the order of the profile equation (1.1), which is one less than the order of the evolution equation: Kuramoto–Sivashinsky is a fourth-order equation and has $p = 3$, Kawahara has $p = 4$, and the seventh-order KdV equation has $p = 6$. Phrases such as “order three” or “in every order” always refer to p .

In the evolution equation, derivatives of odd order are dispersive terms and derivatives of even order are dissipative terms (in this sign convention a positive coefficient of u_{xx} is gain, so dissipative includes gain). We call an equation *purely dispersive* if it contains only odd-order derivatives, as the KdV and Kawahara equations do; then p is even and P is an even polynomial. We call it *purely dissipative* if its linear part contains only even-order derivatives, as for the Kuramoto–Sivashinsky equation without dispersion; then p is odd and $P - P(0)$ is an odd polynomial. All other equations, such as KdV–Burgers, are *mixed*. The first two classes are *the two pure subfamilies*; each order p has one of them, according to its parity.

The waves. A travelling wave is *meromorphic* if its profile is a single-valued meromorphic function of the complex variable z . The familiar exact solutions are of this kind: a polynomial in $\tanh$ or sech is a rational function of an exponential $e^{\kappa z}$, a cnoidal wave is an elliptic function, and rational functions of z also occur. These are the three kinds of wave in this paper. We consider those with *one pole per period*: all the poles of the profile are translates of a single pole by its periods. For a wave of $\tanh$ type the period in question is the imaginary period $2\pi i/\kappa$ of $e^{\kappa z}$, and the rate κ measures the width of the wave. Such a wave tends to a constant at each end (Section 2.3); it is a *pulse* if it returns to the same constant, and a *front* if it connects two different constants.

For every equation of even order, and for every purely dispersive equation, these are all the meromorphic travelling waves that exist (Theorem 5.1). For the remaining equations, the mixed ones of odd order such as KdV–Burgers, other meromorphic waves can occur; there we count the one-pole waves without claiming more. These equations are the rows marked † in Table 1.

The question. For $p \geq 3$ such waves are exceptional: an equation of the family has one only if its coefficients satisfy algebraic relations, as the four values of σ illustrate. This suggests the right question. We do not ask how many waves a given equation has, but how many pairs of an equation and a wave there are in the whole family.

To make this question finite, two continuous freedoms must be removed. A translate of a wave is again a wave, so we place the pole at a fixed point. Rescaling x changes the width of a wave and the coefficients of its equation together, so we fix the width: for a wave that is rational in an exponential we prescribe the rate $\kappa = 1$, and for an elliptic wave we prescribe both of its periods,

that is, its period lattice. The coefficients of the equation remain free. We therefore count pairs (P, v) , where P ranges over all monic polynomials of degree p and v solves (1.1):

- a *rational-exponential pair* has v rational in e^z , with one pole modulo $2\pi i\mathbb{Z}$, placed at $i\pi$, so that the profile is smooth on the real axis, and $v \rightarrow 0$ as $e^z \rightarrow 0$, that is, as $z \rightarrow -\infty$ along the real axis;
- an *elliptic pair* has v elliptic on a given lattice, with its only pole at zero.

A wave read in the opposite direction, $z \mapsto -z$, is counted separately. At the same rate it is a wave of the mirror-image equation, in which the terms of one parity have changed sign, and restoring the normalization gives again a pair, the *mirror image* of the first. For a fixed equation it is the same wave, read with the rate $-\kappa$.

One more convention explains a factor two below. The profile equation has two constant solutions, and a wave can be measured from either of them. The change is a Galilean shift: it changes the speed relative to the background, and with it P , but not the wave (Section 2). A rational-exponential wave is measured from its limit as $e^z \rightarrow 0$. An elliptic wave has no such limit, and both choices are counted, so every elliptic wave appears in two pairs.

The answer. Four numbers appear:

$$\begin{aligned} N_p &= \binom{2p-1}{p-1} && \text{rational-exponential pairs,} \\ S_p &= \binom{2p-2}{p-2} && \text{the pulses among them; the other } N_p - S_p \text{ are fronts,} \\ M_p &= 2S_p && \text{elliptic pairs on a generic lattice,} \\ F_p &= \binom{p-1}{\lfloor (p-1)/2 \rfloor} && \text{rational-exponential pairs of the pure subfamily of order } p. \end{aligned}$$

We call $2p-1$ the *index* of the order p . Our main result establishes these numbers whenever the index is a prime number:

- there are exactly N_p rational-exponential pairs, and they are all distinct (Theorem 3.4);
- exactly S_p of them are pulses, and the others are fronts (Proposition 3.6);
- a generic lattice carries exactly $M_p = 2S_p$ elliptic pairs, and they are all distinct (Theorem 4.1); generic means every lattice except those with finitely many values of the invariant $j(\Lambda)$, which at $p=3$ are $j=0$ and -300 ;
- exactly F_p of the rational-exponential pairs belong to the pure subfamily of order p , and they are the pairs that coincide with their mirror image. Those of a purely dispersive equation are pulses, and those of a purely dissipative one are fronts. For even p , $2F_p$ of the elliptic pairs belong to purely dispersive equations (Theorem 6.2, Corollary 6.4).

The index is prime for $p = 2, 3, 4, 6, 7, 9, 10, 12, \dots$. At the three remaining orders up to twelve, $p = 5, 8$ and 11 , the same statements are established by exact computer calculations (Corollary 9.2 and Proposition 9.1(iii)). We call the orders with a prime index, together with all $p \leq 12$, the *covered* orders; they include every named equation of Table 1. We conjecture the same statements in every order (Conjecture 9.3). At a prime index the first statement has also been verified formally, in Lean (Remark 3.5).

Table 1: The named equations of the family with $p \leq 6$; here $u_{kx} = \partial_x^k u$. Each name stands for a family with free coefficients, and the numbers count distinct pairs across that family: rational-exponential pairs at unit rate, and elliptic pairs on a generic given lattice, where every elliptic wave appears twice. Except in the two rows marked $\dagger$, these pairs are all the nonconstant meromorphic travelling waves of the equations, apart from a rational wave when all lower coefficients vanish.

Equation	Linear terms	p	Rational-exponential pairs	Elliptic pairs
KdV	u_{3x}	2	$F_2 = 1$	$2F_2 = 2$
KdV–Burgers †	u_{2x}, u_{3x}	2	$N_2 = 3$	$M_2 = 2$
Kuramoto–Sivashinsky (KS)	u_{2x}, u_{4x}	3	$F_3 = 2$	0
KS with dispersion	u_{2x}, u_{3x}, u_{4x}	3	$N_3 = 10$	$M_3 = 8$
Kawahara	u_{3x}, u_{5x}	4	$F_4 = 3$	$2F_4 = 6$
Benney–Lin †	$u_{2x}, \dots, u_{5x}$	4	$N_4 = 35$	$M_4 = 30$
Nikolaevskiy	u_{2x}, u_{4x}, u_{6x}	5	$F_5 = 6$	0
Nikolaevskiy with dispersion	$u_{2x}, \dots, u_{6x}$	5	$N_5 = 126$	$M_5 = 112$
seventh-order KdV	u_{3x}, u_{5x}, u_{7x}	6	$F_6 = 10$	$2F_6 = 20$

The named equations of Table 1 are the Benney–Lin equation [3, 62], the Kuramoto–Sivashinsky equation with dispersion, also called the generalized Kuramoto–Sivashinsky, Benney or KdV–KS equation [28], the Nikolaevskiy equation [68] and its dispersive version [40, 71], the long-wave equations for a liquid with gas bubbles [51, 52] and for flow in a viscoelastic tube [39, 43], where the mixed fifth-order equation of the Benney–Lin row was also derived [53] and studied numerically [54], and the equations already mentioned. The rows with N_p and M_p are full families, and a full family contains the pure subfamily of the same p . The small entries are familiar. The three pairs of KdV–Burgers are the KdV soliton and the KdV–Burgers front, read in its two directions. The two elliptic pairs of KdV are the cnoidal wave on its two backgrounds. The eight elliptic pairs at $p = 3$ are Kudryashov’s elliptic solution: on a given lattice it occurs four times, always with $\sigma = \pm 4$, and each time on two backgrounds (Section 8.2). The two entries 0 say that these purely dissipative equations have no elliptic pair on a generic lattice (Section 6); at $p = 3$ they have one only when $g_2 = 0$. The six pairs of the purely dissipative family with $p = 5$ are the six solitary waves of Kudryashov and Demina [48], four of them real [40] (Section 8.4).

Some remarks make precise what the numbers mean. The counts are over $\mathbb{C}$ and range over all monic P . Many of the pairs have real coefficients and give smooth bounded waves of real equations: eight of the ten at $p = 3$ (Table 2), and 21 of the 35 at $p = 4$ (Section 8.3). A pair with non-real coefficients at unit rate can still belong to a real equation, at an imaginary rate; the wave is then periodic in x and has poles on the real axis (Section 3.2). The word *distinct* carries content: a solution of a polynomial system can be multiple, like a multiple root of a polynomial, and on special lattices elliptic pairs do merge in this way (Section 8.2). Rational waves are not counted: at the covered orders the only one with a single pole is $v = c_* z^{-p}$, for $P = D^p$ [44]; the constant c_* is given in (2.7). Finally, the waves of low order are classical. The general enumeration has no precedent that we know of. Its special cases at $p = 3$, in the Kawahara family [18, 32] and in the purely dissipative family with $p = 5$ [48] are in the literature, as Sections 3.2 and 8 make explicit; what is new is that the lists are complete, and how long they are. We do not discuss stability, and we treat only the quadratic nonlinearity uu_x .

A single equation. The numbers in Table 1 are totals over all coefficient values of a family. An individual equation has such a wave only if its coefficient ratios lie on special curves (Section 7). The reason is that rescaling x moves an equation along a curve in the space of coefficient ratios, and each pair marks one curve. In a family with a single coefficient ratio (KdV–Burgers, Kawahara, Kuramoto–Sivashinsky without dispersion) a rescaling carries every equation with a nonzero ratio to every other one. Each of these equations then has all the waves whose ratio is not zero: the KdV–Burgers front, and all $F_4 = 3$ and $F_3 = 2$ wave shapes of the other two families. The rescaling may be complex: in the last two families a real one cannot change the sign of the ratio. An equation without lower-order terms, such as KdV, is invariant under rescaling and has each of its waves at every rate.

Whenever $2p - 1$ is prime, and for every $p \leq 573$, the equations with a one-pole rational-exponential wave are those on finitely many curves, at most one for each pair and its mirror image (Corollary 7.2). This is the precise sense in which the waves are exceptional for $p \geq 3$. At $p = 3$ the curves are $b_2^2 = \sigma^2 b_1 b_3$ with the four values of σ . For equations of even order the classification gives more: a generic such equation has no nonconstant meromorphic travelling wave at all, at the same orders (Corollary 7.3, and the remark after it for $p = 3$); generic means outside a proper algebraic subset of the coefficients. It may well have travelling waves; but none of them is meromorphic. Section 7 also records two consequences for the way exact travelling waves are found in practice. The counts apply to the many other equations, with mixed derivatives or in several space variables, that have the same profile equation. And they are a checklist for the output of the expansion methods in powers of $\tanh$ or of a Jacobi function: always for the equations of Theorem 5.1, and otherwise for outputs with one pole per period.

When these are all the meromorphic waves. For every equation of even order (p odd) and for every purely dispersive equation, the one-pole waves of the three kinds are all the nonconstant meromorphic travelling waves there are (Theorem 5.1). The proof rests on Eremenko’s theorem [22]: an autonomous algebraic differential equation has only these three kinds of meromorphic solutions if it has a single monomial of top degree, here $v^2/2$, and finitely many Laurent expansions at a pole. For odd p the Laurent expansion at a pole is unique. For a purely dispersive equation one Laurent coefficient is free, a resonance; but the equation conserves an energy, and fixing the energy fixes that coefficient. For a given operator P , which records the equation and the speed, there is therefore at most one such profile up to translation when p is odd, and at most one on each energy level when the equation is purely dispersive. A by-product is that every wave of a pure subfamily is symmetric about its poles, or antisymmetric up to a constant (Corollary 5.3). The theorem is a uniform version of results that exist in pieces: the case of odd p is contained in Yuan et al. [74, Theorem 1], and the energy device is that of Eremenko et al. [23, Lemmas 3–4]; Section 5 gives the detailed attributions. The polynomial systems for the three kinds of wave are instances of the principal-part construction of Demina and Kudryashov [19, 20]. The mixed equations of odd order are not included, and cannot be: the profile equation of KdV–Burgers is also that of the Fisher equation, and at the speed of Ablowitz and Zeppetella it has meromorphic solutions of none of the three kinds (Remark 5.2).

Every wave has a label. Here is where the binomial coefficient N_p comes from. Put $Q = 1/(1 + e^{-z}) = (1 + \tanh(z/2))/2$. Every rational-exponential pair has a profile of the form $v_A = s_p A(D)Q$, where $s_p = 2(2p - 1)!/((p - 1)!)^2$ is a fixed scale ($s_2 = 12$, $s_3 = 60$) and A is a monic polynomial of degree $p - 1$; this is a polynomial in $\tanh(z/2)$ written in a convenient basis. For $\operatorname{Re} z < 0$ we have $Q = e^z - e^{2z} + e^{3z} - \dots$, and D multiplies e^{nz} by n . Hence the

coefficient of e^{nz} is $\pm s_p A(n)$ in v_A and $\pm s_p P(n)A(n)$ in $P(D)v_A$, while in v_A^2 it is $\pm s_p^2 S_A(n)$ with the convolution sum

$$S_A(n) = \sum_{j=1}^{n-1} A(j)A(n-j).$$

These sums are the values of a polynomial S_A of degree $2p-1$, and (1.1) holds exactly when $2P(n)A(n) = s_p S_A(n)$ for all n . So the pair exists if and only if the polynomial A divides the polynomial S_A , and then P is the quotient, up to the constant factor (Proposition 3.1).

The smallest case shows what happens. For $p=2$ and $A = \rho + a$,

$$S_A(\rho) = (\rho-1)\left(\frac{\rho(\rho+1)}{6} + a\rho + a^2\right),$$

which is divisible by $\rho + a$ exactly when $a(a-1)(a+1) = 0$. The solution $a = 0$ is the KdV soliton, and $a = \pm 1$ are the KdV–Burgers front read in its two directions. Now reduce modulo $3 = 2p-1$. Whatever the value of a ,

$$3S_A(\rho) \equiv -(\rho^3 - \rho) = -\rho(\rho-1)(\rho+1) \pmod{3},$$

so modulo 3 the divisor A must be one of the three factors ρ , $\rho-1$, $\rho+1$. Each of them comes from exactly one solution, and the root of A modulo 3 serves as a label: $\{0\}$ for the soliton, $\{2\}$ for the front $a = 1$, that is, the root $-1 \equiv 2$, and $\{1\}$ for the front $a = -1$.

The same happens whenever $\ell = 2p-1$ is prime. The polynomial S_A is a combination of the binomial coefficients $\binom{\rho-1}{k}$, and only the last of them, with $k = \ell$, has ℓ in its denominator. For this reason ℓS_A reduces modulo ℓ to $\pm(\rho^\ell - \rho)$, whatever A is, and the condition becomes that A divides $\rho^\ell - \rho$ modulo ℓ . Now $\rho^\ell - \rho$ is the product of $\rho - s$ over all residues s modulo ℓ . Its monic divisors of degree $p-1$ are therefore given by the $(p-1)$ -element subsets L of the integers modulo ℓ , and there are $\binom{\ell}{p-1} = N_p$ of them. We prove that every pair can be reduced modulo ℓ , although its coefficients are in general algebraic numbers and not integers, and that every divisor comes from exactly one pair. An algebraic number is reduced through an equation that it satisfies: i is 2 or 3 modulo 5, because $2^2 \equiv 3^2 \equiv -1$. For instance $A = \rho^2 + 1/11$, the Kuramoto–Tsuzuki wave, reduces to $\rho^2 + 1 = (\rho-2)(\rho-3)$ modulo 5, and has the label $\{2, 3\}$.

So each pair has a label L , and the labels carry information. The pair is a pulse exactly when L contains 0. At $p=3$ the ten classical waves correspond to the ten two-element subsets of $\{0, 1, 2, 3, 4\}$ (Table 2), and the four pulses are the subsets that contain 0. The mirror image of the pair with label L has label $-L$. The pairs with $L = -L$ coincide with their mirror image, and they are exactly the pairs of the two pure subfamilies. There are F_p such subsets.

The elliptic count comes from the pulses. When one period of the lattice tends to infinity, an elliptic pair becomes a pulse, measured from one of its two constant backgrounds. We show that every pulse, on either background, arises in this way exactly once, and this gives $M_p = 2S_p$.

Other orders. The number N_p is also the largest number of isolated solutions, counted with multiplicity, that matching equations of this shape can have (Proposition 3.2). It is attained if no solution lies at infinity, which is rigidity, and the solutions are distinct if all of them are simple. These are the two facts that are not yet proved at every order. In terms of waves, rigidity says that the rational wave $c_* z^{-p}$ above is the only one with a single pole (Lemma 3.3). It is proved when $2p-1$ is a prime or a prime power, and established by exact computations for $p \leq 573$. Simplicity is proved when $2p-1$ is prime and, by exact computations, at $p = 5, 8$ and 11 . Section 9 states what follows from each of them; the proofs are in a companion note in the source package.

Organization. Section 2 contains the normalizations and the form of the waves. The counts come first (Sections 3 and 4), because they do not use the classification. Their proofs, in Section 3.3 and in Section 4, can be skipped; every later section uses only the statements. Then come the classification (Section 5), the two pure subfamilies (Section 6), a single equation (Section 7), the examples (Section 8), and the other orders with the open questions (Section 9). Appendix A describes the exact verification and the ancillary material.

2 Equations, waves and what is counted

This section fixes the normalizations and the vocabulary, and records the form of the waves that are counted. The normalizations are what makes a finite count possible, and they are also what connects the count with a given evolution equation (Section 7).

2.1 From the evolution equation to the profile equation

Let $u = w(\xi)$, $\xi = x - ct$, be a travelling wave of

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0. \quad (2.1)$$

One integration gives

$$\sum_{j=1}^p b_j w^{(j)} - cw + \frac{1}{2}w^2 + K = 0 \quad (2.2)$$

with a constant K . Let w_* be a constant solution of (2.2), that is, a root of $w^2/2 - cw + K$, and let $\kappa \neq 0$. The substitution

$$w(\xi) = w_* + b_p \kappa^p v(\kappa \xi) \quad (2.3)$$

turns (2.2) into the profile equation (1.1) with

$$a_j = \frac{b_j}{b_p} \kappa^{j-p} \quad (1 \leq j \leq p-1), \quad a_0 = \frac{w_* - c}{b_p \kappa^p}. \quad (2.4)$$

Thus $a_1, \dots, a_{p-1}$ record the equation, up to the scale κ , and a_0 records the speed of the wave relative to the background w_* . The scale is free at this stage. It will be fixed by the period of the wave: if v is rational in e^z , then w is rational in $e^{\kappa \xi}$, so that κ is the rate of the wave.

The constant solutions of (1.1) are $v = 0$ and $v = -2a_0$. They correspond to the two possible choices of the background w_* . Changing that choice replaces the pair (P, v) by

$$(P - 2P(0), v + 2P(0)), \quad (2.5)$$

which again satisfies (1.1), as a direct substitution shows. In the evolution equation the change of background is a Galilean shift $u \mapsto u + k$, $x \mapsto x - kt$; it changes the speed relative to the background, hence the sign of a_0 . The two pairs describe the same solution $u(x, t)$ of the same evolution equation. They coincide exactly when $a_0 = 0$. For example, the KdV soliton gives the pair $P = \rho^2 - 1$, $v = 3 \operatorname{sech}^2(z/2)$. Measured from the other background it is $P = \rho^2 + 1$ and $v - 2$, which does not tend to zero and is therefore not a rational-exponential pair.

2.2 Dispersive, dissipative and mixed equations

In (2.1) the term $b_j \partial_x^{j+1} u$ is dispersive for even j and dissipative for odd j : for real coefficients it changes the phase, respectively the amplitude, of a Fourier mode (with the signs of (2.1), a positive coefficient of u_{xx} is gain). The equation is *purely dispersive* if $b_j = 0$ for odd j . Then p is even, P is an even polynomial, and the equation is invariant under $(x, t) \mapsto (-x, -t)$. The equation is *purely dissipative* if $b_j = 0$ for even j . Then p is odd, $P - P(0)$ is an odd polynomial, and the equation is invariant under $(x, u) \mapsto (-x, -u)$; the Nikolaevskiy equation ($p = 5$) is an example, see Simbawa et al. [71]. All other equations are *mixed*.

The two pure subfamilies are described by one condition. We call a monic polynomial P of degree p *reflection-symmetric* if $P - P(0)$ has the parity of ρ^p :

$$P(-\rho) = P(\rho) \quad (p \text{ even}), \quad P(\rho) + P(-\rho) = 2P(0) \quad (p \text{ odd}). \quad (2.6)$$

In the travelling-wave variable both symmetries are the reflection $z \mapsto -z$. If P is even and $v(z)$ solves (1.1), so does $v(-z)$. If p is odd and $P - P(0)$ is odd, so does $-v(-z) - 2P(0)$; this is the reflection combined with the change of background (2.5).

2.3 Poles, and the form of one-pole waves

To count waves we need to know what they look like. Three facts suffice: a pole has order p and a fixed leading coefficient; a wave rational in e^z with one pole per period is a polynomial of degree p in the logistic function Q ; an elliptic wave with one pole per period is a combination of $\wp$ and its derivatives.

Poles. Throughout, a meromorphic function is single-valued and meromorphic on the whole complex plane. The order of a pole of a solution of (1.1) is forced by dominant balance. A pole of order r makes $P(D)v$ of order $r + p$ and v^2 of order $2r$; the two can cancel only if $r = p$. Comparing the leading coefficients gives, at a pole z_0 ,

$$v(z) = \frac{c_*}{(z - z_0)^p} + O((z - z_0)^{1-p}), \quad c_* = (-1)^{p-1}(p-1)! s_p, \quad s_p = \frac{2(2p-1)!}{((p-1)!)^2}. \quad (2.7)$$

This holds for every P . The constant s_p is the scale of the normal forms of Sections 3 and 4; its first values are $s_2 = 12$, $s_3 = 60$, $s_4 = 280$, $s_6 = 5544$, so that $c_* = -12, 120, -1680$ for $p = 2, 3, 4$.

Three kinds of wave. The meromorphic travelling waves in this paper are of three kinds: rational in z , rational in an exponential $e^{\kappa z}$, and elliptic. For evolution equations of even order and for purely dispersive ones there are no others (Theorem 5.1); for the remaining equations these are the waves that we count. The period group of such a wave is its *full* group of periods: trivial in the first case, generated by one period in the second, a lattice in the third. In the second case κ is chosen so that $2\pi i/\kappa$ generates the period group. The wave has *one pole per period* if all its poles are translates of a single pole by the periods. A rational wave then has a single pole, and an elliptic wave has one pole in each period parallelogram.

Waves rational in an exponential. Let $v = R(t)$, $t = e^{\kappa z}$, be such a wave with one pole per period. The rational function R has finite limits at both $t = 0$ and $t = \infty$. Indeed, $P(D) = P(\kappa t \partial_t)$ multiplies each monomial t^m by a constant, whereas squaring doubles an extreme exponent: if R

had a pole at either end, R^2 would contain an extreme Laurent monomial that $P(\kappa t \partial_t)R$ cannot cancel. Both limits are constant solutions of (1.1). After a translation, the only pole of R in $0 < |t| < \infty$ is at $t = -1$, of order p . For $\kappa = 1$ such a function is a polynomial of degree p in the logistic function

$$Q = \frac{1}{1 + e^{-z}} = \frac{1 + \tanh(z/2)}{2}, \quad DQ = Q(1 - Q),$$

which has a simple pole with residue one at $z = i\pi$ and the limits 0 and 1. By (2.7) the coefficient of Q^p is c_* ; the KdV soliton above is $12Q(1 - Q)$. Polynomial logistic representations and their use in constructing travelling waves are developed by Kudryashov [38, 55, 56], and applied to equations of the family up to $p = 6$ by Ryabov et al. [70].

The wave is a *pulse* if its two limits coincide and a *front* otherwise. For a real rate and a real profile these are the solitary waves and the kinks of the evolution equation, and they are smooth and bounded on the real axis when the pole lies off it. In (2.3) we take w_* to be the limit as $e^{\kappa z} \rightarrow 0$. Then $v \rightarrow 0$ at that end, and the other limit is 0 for a pulse and $-2a_0$ for a front.

Elliptic and rational waves. An elliptic wave whose only pole modulo its periods is at zero is a constant plus a linear combination of $\wp, \wp', \dots, \wp^{(p-2)}$, in which $\wp^{(p-2)}$ has the coefficient $-s_p$ (Section 4). Rational waves are not counted, but they govern what happens at infinity in both counts. A rational solution has no nonconstant polynomial part at infinity: such a part would have its degree doubled by v^2 , while the linear differential expression cannot reach that degree. A rational wave whose only pole is at zero is therefore

$$v(z) = c_* z^{-p} + (\text{a polynomial of degree at most } p - 1 \text{ in } z^{-1}). \quad (2.8)$$

The simplest one is $v = c_* z^{-p}$, for $P = D^p$; it is given for every order by Kudryashov [44].

Conversely, a function of one of these three forms that satisfies (1.1) is a nonconstant wave with one pole per period, because $c_* \neq 0$. Substitution turns (1.1) into finitely many polynomial equations in the coefficients of the form and of P , without any genericity assumption. These equations are instances of the principal-part construction of Demina and Kudryashov [19, 20]; for a fixed operator the same construction reduces to the coefficient comparison of Demina and Kudryashov [14], which we do not need.

2.4 What is counted

We count pairs (P, v) of a monic polynomial P of degree p and a solution v of (1.1). In a *rational-exponential pair*, v is rational in e^z , its poles are the points of $i\pi + 2\pi i\mathbb{Z}$, and $v \rightarrow 0$ as $e^z \rightarrow 0$. In an *elliptic pair* on a lattice Λ , v is elliptic with period lattice Λ , including its scale, and its poles are the points of Λ . Fixing the periods removes the scale κ in (2.3), and placing the pole removes translations. The operator P varies over *all* monic polynomials of degree p , and all counts are over $\mathbb{C}$. In both cases a wave and its mirror image are counted as two pairs when they are distinct (Section 6). Both pairs (2.5) of an elliptic wave are counted; for a rational-exponential wave the condition $v \rightarrow 0$ selects one of them.

3 The count of rational-exponential pairs

We count the pairs (P, v) of Section 2 for all monic P of degree p . Whether an equation can have meromorphic waves other than these is a separate question; Section 5 answers it, and nothing in this section or the next depends on the answer.

By Section 2.3, a normalized profile is a polynomial of degree p in $Q = 1/(1 + e^{-z})$, without constant term because $v \rightarrow 0$ as $e^z \rightarrow 0$. The functions $Q, DQ, \dots, D^{p-1}Q$ span the degree- p polynomials in Q with zero constant term, since $D^j Q = (-1)^j j! Q^{j+1}$ plus lower powers of Q . The leading pole coefficient (2.7) therefore gives the unique form

$$v_A = s_p A(D)Q, \quad A(\rho) = \rho^d + \alpha_1 \rho^{d-1} + \dots + \alpha_d, \quad d = p - 1, \quad (3.1)$$

with the scale s_p of (2.7). This representation is the standard singular-part operator form: $Q = 1/2 + D \log \cosh(z/2)$, so (3.1) is a constant plus a linear differential operator applied to $\log \cosh(z/2)$; such forms, including the KS case, are discussed by Conte and Musette [5]. We use $\alpha_1, \dots, \alpha_d$ as the unknowns of the count.

3.1 Discrete convolution and the matching equations

For polynomials F, G , define $F \star G$ by polynomial continuation of

$$(F \star G)(n) = \sum_{j=1}^{n-1} F(j)G(n-j), \quad n \geq 1.$$

The continuation exists by the power-sum formulas: for fixed F, G the right side is a polynomial in n . Write $S_A = A \star A$. For monic A of degree d ,

$$\deg S_A = 2d + 1, \quad [\rho^{2d+1}]S_A = \frac{(d!)^2}{(2d+1)!} = \frac{2}{s_p}. \quad (3.2)$$

The leading coefficient is the beta integral $\int_0^1 t^d(1-t)^d dt$.

Proposition 3.1 (Polynomial divisibility). *The profile (3.1) solves (1.1) for some monic P if and only if A divides S_A . The operator is unique and is*

$$P_A = \frac{s_p S_A}{2A}. \quad (3.3)$$

Proof. In the half-plane $\operatorname{Re} z < 0$, the logistic function is a geometric series in e^z , and applying $A(D)$ multiplies the n th term by $A(n)$:

$$A(D)Q = \sum_{n \geq 1} (-1)^{n-1} A(n) e^{nz}.$$

Squaring gives the rational-function identity

$$(A(D)Q)^2 = -S_A(D)Q. \quad (3.4)$$

The map $F \mapsto F(D)Q$ is injective: its vanishing would give $F(n) = 0$ for every positive integer n . Consequently (1.1) is equivalent to $2P_A A = s_p S_A$, which has a polynomial solution P_A exactly when A divides S_A . Equation (3.2) shows that P_A is monic of degree p . $\square$

Divide S_A by the monic polynomial A :

$$S_A = AC_A + R_A, \quad R_A = \sum_{r=0}^{d-1} R_r(\alpha_1, \dots, \alpha_d) \rho^r. \quad (3.5)$$

The matching equations are $R_0 = \dots = R_{d-1} = 0$: d polynomial equations in the d unknowns $\alpha_1, \dots, \alpha_d$. Their solutions are in bijection with the normalized pairs (P_A, v_A) . For $p = 2$ and $A = \rho + a$ there is the single equation $R_0 = -a(a-1)(a+1)/6 = 0$ of the Introduction. We write $\mathcal{X}_p$ for the set of solutions of the matching equations. The multiplicity of a point of $\mathcal{X}_p$ always means its multiplicity as a solution of these equations, in the following sense.

Simple solutions and multiplicity. The solutions of a system of d polynomial equations in d unknowns are counted in the way the roots of a polynomial are counted. A solution is *simple* if the Jacobian matrix of $(R_0, \dots, R_{d-1})$ with respect to $(\alpha_1, \dots, \alpha_d)$ is invertible there; this generalizes a simple root, $f(a) = 0$, $f'(a) \neq 0$. Every isolated solution, that is, one that is not a limit of other solutions, has a *multiplicity*, a positive integer that generalizes the multiplicity of a root: it is the number of solutions near it of the perturbed system $R_r = \varepsilon_r$, for generic small $\varepsilon \in \mathbb{C}^d$, in the way in which the double root of $x^2 = 0$ becomes the two roots of $x^2 = \varepsilon$. It is one exactly when the solution is simple. Computer algebra obtains it as a dimension. The multiplicity of a solution is the dimension over $\mathbb{C}$ of the space of convergent power series at that point modulo the combinations $\sum_r h_r R_r$ with power series h_r ; and when a system has finitely many solutions, the sum of their multiplicities is the dimension of the space of polynomials modulo the combinations $\sum_r h_r R_r$ with polynomials h_r [2, Sections 4.3 and 5.2], [10, Chapter 4, §2]. A *count with multiplicity* is this sum. It equals the number of distinct pairs exactly when every solution is simple. Section 8.2 shows multiplicities at work.

How many solutions can there be? Before counting, we ask how many solutions such a system can have at most. The answer is already N_p . Give α_j weight j and ρ weight one, so that A is weighted-homogeneous of weight d . This grading reflects how the coefficients scale: replacing the roots of A by λ times themselves multiplies α_j by λ^j . The polynomial S_A has weight at most $2d + 1$, and its part of weight exactly $2d + 1$ is the *continuous* convolution

$$H_A(\rho) = \int_0^\rho A(t)A(\rho - t) dt, \quad \int_0^\rho t^i(\rho - t)^j dt = \frac{i!j!}{(i+j+1)!} \rho^{i+j+1}. \quad (3.6)$$

Indeed these are precisely the highest weighted terms of the power sums that define S_A . Monic division preserves the highest weighted part: A is weighted-homogeneous and monic, so each step of long division subtracts a multiple of A of the same weight as the term being removed. Consequently the matching equation R_r has weight at most $2d + 1 - r$, and

$$\text{the part of weight } 2d + 1 - r \text{ of } R_r \text{ is the coefficient of } \rho^r \text{ in } \text{rem}_A H_A. \quad (3.7)$$

In words, the discrete convolution S_A is a lower-order perturbation of the continuous convolution H_A , and far from the origin the matching equations look like the continuous problem $A \mid H_A$. Solutions of the matching equations could escape to infinity only along solutions of that problem. We denote by (R_d) the statement that it has none except the trivial one:

$$A \text{ monic, } \deg A = d, \quad A \mid H_A \implies A(\rho) = \rho^d. \quad (3.8)$$

We call (R_d) *rigidity*.

Proposition 3.2 (Weighted Bézout bound). *Let $p \geq 2$ and $d = p - 1$. The matching equations are d equations of weighted degrees at most $2d + 1, 2d, \dots, d + 2$ in d unknowns of weights $1, 2, \dots, d$. Their isolated solutions, counted with multiplicity, number at most*

$$\frac{(d+2)(d+3) \cdots (2d+1)}{d!} = \binom{2d+1}{d} = \binom{2p-1}{p-1} = N_p.$$

If rigidity (R_d) holds, that is, if the parts of weight $2d + 1 - r$ of the R_r , which by (3.7) are the coefficients of $\text{rem}_A H_A$, have no common zero other than the origin, then the matching equations have finitely many solutions, and counted with multiplicity there are exactly N_p of them. They are then N_p distinct solutions exactly when all of them are simple.

For $p = 2$ the bound is $3/1 = 3$, and for $p = 3$ it is $(4 \cdot 5)/(1 \cdot 2) = 10$. The proposition is Bézout's theorem in weighted form (the substitution $\alpha_j = y_j^j$ turns weights into ordinary degrees and counts every solution $d!$ times). It is proved as Theorem 2.1 of the companion note, from Bézout's inequality for isolated solutions [65, Chapter VIII] and Bézout's theorem [10]. The proof of Theorem 3.4 does not use this proposition: at a prime index the solutions are found through their labels, and rigidity comes out as a by-product. The proposition says what the number N_p is, and it is what remains at the other orders (Section 9).

For waves, rigidity is a statement about rational solutions. The dictionary is the Laplace transform, restricted to polynomials.

Lemma 3.3 (Laplace correspondence). *Let $F \mapsto F^\sharp$ be the linear map on polynomials with $(\rho^k)^\sharp = k! z^{-k-1}$. Then $D(F^\sharp) = (-\rho F)^\sharp$, and $F^\sharp G^\sharp = (F * G)^\sharp$ for the continuous convolution $(F * G)(\rho) = \int_0^\rho F(t)G(\rho - t) dt$; in particular $(A^\sharp)^2 = (H_A)^\sharp$. Let P be monic of degree p , and let V be a nonconstant rational solution of (1.1) whose only pole is at $z = 0$. Then V vanishes at infinity, $P(0) = 0$, and*

$$V = (-1)^{p-1} s_p A^\sharp, \quad A \mid H_A, \quad P(\rho) = (-1)^p \frac{s_p}{2} \frac{H_A}{A}(-\rho),$$

with A monic of degree $p - 1$. Conversely, every monic A of degree $p - 1$ that divides H_A gives such a solution. The residue of V at zero is $(-1)^{p-1} s_p A(0)$.

Proof. Both rules are checked on monomials; the second is the integral in (3.6). By (2.8), $V = b + f$ with a constant b and with $f \rightarrow 0$ at infinity. The constant part of (1.1) is $b(P(0) + b/2) = 0$, and the slowest decaying term of f enters the rest of the equation with the factor $P(0) + b$, which must vanish. Hence $b = P(0) = 0$. By (2.7) the pole has order p and leading coefficient $c_* = (-1)^{p-1} (p-1)! s_p$, so $V = cA^\sharp$ with $c = (-1)^{p-1} s_p$ and A monic of degree $p - 1$. By the two rules, (1.1) for V reads $(cP(-\rho)A + c^2 H_A/2)^\sharp = 0$, and the map $\sharp$ is injective. This gives the divisibility and the formula for P . It also gives the converse; the polynomial P defined by the formula is monic of degree p , because H_A has the leading coefficient (3.2) of S_A . The residue is the coefficient $cA(0)$ of z^{-1} . $\square$

For $A = \rho^{p-1}$ the lemma gives $V = c_* z^{-p}$ and $P = D^p$. Rigidity (R_{p-1}) is therefore the statement that, for every monic P of degree p , equation (1.1) has no other nonconstant rational solution with a single pole.

3.2 The prime-index theorem

Theorem 3.4 (Prime-index enumeration). *Let $\ell = 2p - 1$ be prime, and put $d = p - 1$.*

- (a) *The matching equations have exactly $\binom{\ell}{d} = N_p$ solutions over $\mathbb{C}$, and every solution is simple. Thus there are exactly N_p rational-exponential pairs, and they are distinct.*
- (b) *The coefficients of every solution are algebraic numbers, and they are ℓ -integral: they have no ℓ in a denominator and can be reduced modulo ℓ , in the way in which $1/11$ becomes 1 and i becomes 2 or 3 modulo 5. Once a consistent way of reducing algebraic numbers is fixed (Section 3.3), reduction modulo ℓ is a bijection between the solutions and the d -element subsets L of the field $\mathbb{F}_\ell$ of integers modulo ℓ :*

$$\overline{A}(\rho) = \prod_{s \in L} (\rho - s). \tag{3.9}$$

(c) *Rigidity* (R_d) *holds*.

We call L the *label* of the pair. It depends on the way of reducing algebraic numbers, an embedding that is fixed in the proof, only in the way in which i is 2 or 3 modulo 5: another embedding gives a solution the label that the first one gives to a conjugate solution. Whether a label contains 0, and the labels of solutions with rational coefficients, do not depend on it. The labels describe reductions of algebraic coefficients; they are not assertions that the complex roots of A , or the characteristic roots of P_A , are integers.

Example: order three. Let $p = 3$, $s_3 = 60$, and $A(\rho) = \rho^2 + a\rho + b$, so that $(a, b) = (\alpha_1, \alpha_2)$. The profile and monic operator are

$$\frac{v_A}{60} = 2Q^3 - (a + 3)Q^2 + (a + b + 1)Q, \quad (3.10)$$

$$P_A(\rho) = \rho^3 + 4a\rho^2 + (a^2 + 19b)\rho - a^3 + 7ab - 5a - 30b. \quad (3.11)$$

The remainder of S_A is $(\hat{R}_1\rho + \hat{R}_0)/30$, where

$$\hat{R}_0 = 30R_0 = ab(a^2 - 7b + 5), \quad \hat{R}_1 = 30R_1 = a^4 - 8a^2b + 11b^2 + 10b - 1. \quad (3.12)$$

The system $\hat{R}_0 = \hat{R}_1 = 0$ is solved by cases on the factors of $\hat{R}_0$. If $b = 0$, then $a^4 = 1$. If $a = 0$, then $(b + 1)(11b - 1) = 0$. In the remaining case $a^2 = 7b - 5$, and the second equation becomes $4(b - 2)(b - 3) = 0$. This exhausts the system and gives Table 2. The Jacobian determinant is nonzero at every point, so all $N_3 = 10$ pairs are simple, as Theorem 3.4(a) states for $2p - 1 = 5$.

Table 2: All ten rational-exponential pairs at order three. The label is the set of roots of A modulo 5 (Theorem 3.4); for the two complex pairs it depends on the choice of i modulo 5. The column $|\sigma|$ refers to the equation of the Introduction, $(b_1, b_2, b_3) = (1, \sigma, 1)$: by (2.4), the coefficients $B = 4a$ of ρ^2 and $\mu = a^2 + 19b$ of ρ in (3.11) are $B = \sigma/\kappa$ and $\mu = 1/\kappa^2$, so $\sigma^2 = B^2/\mu = 16a^2/(a^2 + 19b)$, which does not change when x is rescaled. In the rows $(\pm i, 0)$ and $(0, -1)$, μ is negative, and the rescaling to that equation has an imaginary rate. The determinant is $\det \partial(\hat{R}_0, \hat{R}_1)/\partial(a, b)$. The last column describes v_A on the real axis, at unit rate.

a	b	Labels	$ \sigma $	Determinant	Real profiles
± 1	0	$\{0, 4\}, \{0, 1\}$	4	-24	Pulses
$\pm i$	0	$\{0, 2\}, \{0, 3\}$	4	-16	Not real
0	-1	$\{1, 4\}$	0	144	Front
0	1/11	$\{2, 3\}$	0	576/121	Front
± 3	2	$\{3, 4\}, \{1, 2\}$	$12/\sqrt{47}$	-144	Fronts
± 4	3	$\{2, 4\}, \{1, 3\}$	$16/\sqrt{73}$	384	Fronts

The ten labels are the ten two-element subsets of $\{0, 1, 2, 3, 4\}$; for instance $\rho^2 + 1/11$ reduces to $\rho^2 + 1 = (\rho - 2)(\rho - 3)$. By (3.10), $v_A/60 \rightarrow b$ as $z \rightarrow +\infty$, so a pair is a pulse exactly when $b = 0$. The four pulses are the pairs whose label contains 0; Proposition 3.6 below shows that this is so at every prime index. The map $(a, b) \mapsto (-a, b)$ is the mirror image of the Introduction, and it replaces a label L by $-L$ (Section 6). There are ten complex pairs, eight with real coefficients. All eight are smooth and bounded on the real axis: two pulses and six fronts. For example, $A = (\rho - 1)(\rho - 2)$ gives $v_A = 120Q^3$ and $P_A = (\rho - 3)(\rho - 4)(\rho - 5)$. These are the

classical solutions in hyperbolic functions [31–34, 61]. The complete parameter list is tabulated by Kudryashov and Zargaryan [42, Table 1] and Conte and Musette [6, Table 1]. Their tables contain six parameter cases because they identify the two signs of the dispersion coefficient: the four mirror pairs $(\pm a, b)$ and the two pairs with $a = 0$. Exhaustion of this classical list follows from Eremenko [22]; Demina [16] classifies all irreducible algebraic invariants of the same profile equation. Later rederivations by ansatz methods coincide with these waves [41]. The pairs $(\pm 1, 0)$ and $(\pm i, 0)$ belong to the same equations, those with $\sigma^2 = 16$: by (2.4) the rate is $\kappa = \pm 1$ for the first two and $\kappa = \pm i$ for the others (Section 7). Their rates differ by a factor i , so for a real equation the second wave is periodic in x , and its real form has poles on the real axis.

3.3 Proof of the prime-index theorem

The idea was described in the Introduction. The proof adds two things: every solution can be reduced modulo ℓ , by a dominant balance (Step 3); and every label comes from exactly one solution, by Newton’s method in the form of Hensel’s lemma (Step 4). A reader who accepts the theorem can go to Section 3.4: the rest of the paper uses only its statement, and the proof of Proposition 3.6 uses (3.14) and fact (V) below once more.

Reduction modulo ℓ of algebraic numbers. A rational number can be reduced modulo ℓ if ℓ does not divide its denominator; these numbers form a ring, denoted $\mathbb{Z}_{(\ell)}$. The solutions of the matching equations will turn out to be algebraic numbers, and an algebraic number is reduced through a polynomial equation that it satisfies: i becomes 2 or 3 modulo 5, because $x^2 + 1 \equiv (x - 2)(x - 3)$ modulo 5. The ℓ -adic valuation makes these choices for all algebraic numbers at once, so that reduction respects sums and products, and it tells when a number can be reduced at all. Four standard facts suffice. A useful guide is the analogy with a small parameter: think of ℓ as ε , of the valuation as the order of a quantity in ε , and of reduction modulo ℓ as setting $\varepsilon = 0$.

(V) *Valuation and reduction.* For a nonzero rational number $x = \ell^m x'$, with the numerator and the denominator of x' prime to ℓ , the ℓ -adic valuation is $\text{ord}(x) = m$; put $\text{ord}(0) = \infty$. Then

$$\text{ord}(xy) = \text{ord}(x) + \text{ord}(y), \quad \text{ord}(x + y) \geq \min\{\text{ord}(x), \text{ord}(y)\}, \quad (3.13)$$

and $\mathbb{Z}_{(\ell)}$ consists of the rational numbers with $\text{ord}(x) \geq 0$. Let K be an algebraic closure of the field $\mathbb{Q}_{\ell}$ of ℓ -adic numbers. Then K is algebraically closed, it contains $\mathbb{Q}$, and it carries a function $\text{ord}: K \rightarrow \mathbb{Q} \cup \{\infty\}$ that extends the one above, satisfies (3.13), and is infinite only at 0 [30, Chapter III], [66, Chapter II, Theorem (4.8)]. The field $\overline{\mathbb{Q}} \subset \mathbb{C}$ of all algebraic numbers can be embedded in K . The embedding is not unique: two embeddings differ by an automorphism of $\overline{\mathbb{Q}}$. We fix one embedding, and from now on we regard algebraic numbers as elements of K . An element $x \in K$ is ℓ -integral if $\text{ord}(x) \geq 0$. The ℓ -integral elements form a ring $\mathcal{O}$. Those with $\text{ord}(x) > 0$ form a subset $\mathfrak{m}$ that is closed under addition and under multiplication by elements of $\mathcal{O}$, and an element with $\text{ord}(x) = 0$ has its inverse in $\mathcal{O}$. Hence $\mathbb{k} = \mathcal{O}/\mathfrak{m}$ is a field. It contains $\mathbb{F}_{\ell}$, because $\mathbb{Z}_{(\ell)} \cap \mathfrak{m} = \ell\mathbb{Z}_{(\ell)}$. The map $x \mapsto \bar{x}$ from $\mathcal{O}$ to $\mathbb{k}$ is *reduction modulo ℓ* . It respects sums and products, it is the usual reduction on $\mathbb{Z}_{(\ell)}$, and $\bar{x} = 0$ exactly when $\text{ord}(x) > 0$. Polynomials are reduced coefficient by coefficient. Division by a monic polynomial uses only additions and multiplications of coefficients. Hence it stays within any ring that contains the coefficients, for us $\mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$ or $\mathcal{O}$, and it commutes with the substitution of values for the α_j and with reduction modulo ℓ : the reductions of the quotient and of the remainder are the quotient and

the remainder of the reduced polynomials. A monic polynomial whose roots are ℓ -integral has ℓ -integral coefficients.

(H) *Hensel's lemma.* Let $G = (G_1, \dots, G_d)$ be polynomials in d unknowns with coefficients in $\mathbb{Z}_{(\ell)}$, and let $\bar{a} \in \mathbb{F}_\ell^d$ be a solution of the reduced system at which the Jacobian determinant of the reduced system is not zero. Then there is exactly one $a \in \mathcal{O}^d$ with $G(a) = 0$ and with reduction $\bar{a}$. This is Newton's method: started from any integer vector that reduces to $\bar{a}$, the iteration $a \mapsto a - DG(a)^{-1}G(a)$ converges ℓ -adically. It is the counterpart of a familiar fact: a simple root at $\varepsilon = 0$ continues in exactly one way to a power series in ε . Existence is the multivariable Hensel lemma of Conrad [4] over $\mathbb{Q}_\ell$. For uniqueness, two such solutions have their coordinates in a finite extension of $\mathbb{Q}_\ell$ inside K , which is complete [66, Chapter II, Theorem (4.8)], and the same lemma applies there.

(N) *Algebraic solutions.* A system of polynomial equations with rational coefficients that has a solution in $\mathbb{C}^d$ has a solution in $\overline{\mathbb{Q}}^d$. This is the weak Nullstellensatz [11, Chapter 4, §1].

(T) *Finite systems.* If a system of d polynomial equations in d unknowns with rational coefficients has only finitely many solutions in K^d , then all of them lie in $\overline{\mathbb{Q}}^d$, and the system has the same solutions in $\mathbb{C}^d$, with the Jacobian determinant nonzero at the same ones. Indeed, finiteness over an algebraically closed field means that for each i some nonzero polynomial in x_i alone is a combination of the equations [11, Chapter 5, §3]. To find such a combination, with given bounds on the degrees, is a system of linear equations with rational coefficients. Since it has a solution over K in which the polynomial in x_i is not zero, it has such a solution over $\mathbb{Q}$. That polynomial confines the i th coordinate of every solution, in any field that contains $\mathbb{Q}$, to finitely many algebraic numbers.

Proof of Theorem 3.4. The proof has four steps: a congruence for S_A modulo ℓ ; the solution of the reduced equations; integrality, which shows that every solution can be reduced, and then reduces to one of them; and lifting, which shows that each of them comes from exactly one solution. Solutions are first taken in K^d , with K as in (V); the last step returns to $\mathbb{C}$.

Step 1: the congruence. Induction using $(D - n)Q^n = -nQ^{n+1}$ gives

$$Q^n = \frac{(-1)^{n-1}}{(n-1)!} \prod_{j=1}^{n-1} (D - j)Q.$$

Write $(A(D)Q)^2 = \sum_{n=2}^{2p} \gamma_n Q^n$. Each γ_n is an integer polynomial in the coefficients of A , and $\gamma_{2p} = (d!)^2$. By (3.4),

$$S_A(\rho) = \sum_{n=2}^{2p} \frac{(-1)^n \gamma_n}{(n-1)!} \prod_{j=1}^{n-1} (\rho - j).$$

Only the last denominator, $(2p-1)! = \ell!$, contains ℓ . Consequently ℓS_A has coefficients in $\mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$, and

$$\ell S_A(\rho) \equiv \frac{(d!)^2}{(\ell-1)!} \prod_{j=1}^{\ell} (\rho - j) \equiv (-1)^d (\rho^\ell - \rho) \pmod{\ell}. \quad (3.14)$$

The scalar follows by pairing j with $\ell - j$ in $(\ell-1)!$. The product is $\rho^\ell - \rho$ because, by Fermat's little theorem, every residue s modulo ℓ satisfies $s^\ell = s$. The right side does not depend on A :

this is what makes the prime index special. Taking the parts of weight $\ell = 2d + 1$ on both sides gives the corresponding congruence for the continuous convolution,

$$\ell H_A(\rho) \equiv (-1)^d \rho^\ell \pmod{\ell}. \quad (3.15)$$

Both congruences are between polynomials in ρ and $\alpha_1, \dots, \alpha_d$ with coefficients in $\mathbb{Z}_{(\ell)}$. They may therefore be evaluated at any ℓ -integral values of the α_j and of ρ , and then reduced.

Step 2: the reduced equations. By (V) the scaled remainders ℓR_r have coefficients in $\mathbb{Z}_{(\ell)}$, and by (3.14) their reductions are the coefficients of the remainder of $(-1)^d(\rho^\ell - \rho)$ on division by $\bar{A}$. The reduced equations therefore say that $\bar{A}$ divides $\rho^\ell - \rho$. Now $\rho^\ell - \rho = \prod_{s \in \mathbb{F}_\ell} (\rho - s)$, and polynomials over a field factor uniquely. Over any field that contains $\mathbb{F}_\ell$, in particular over $\mathbb{k}$, the monic divisors of degree d are therefore the $\binom{\ell}{d}$ polynomials (3.9), and their coefficients lie in $\mathbb{F}_\ell$. Each of them is a simple solution of the reduced equations. Write $\rho^\ell - \rho = \bar{A}\bar{C}$. For a variation h with $\deg h < d$, the derivative of the remainder is, up to the nonzero scalar $(-1)^d$,

$$h \mapsto -\text{rem}_{\bar{A}}(\bar{C}h), \quad (3.16)$$

because the dividend in (3.14) does not vary with A . Since $\rho^\ell - \rho$ has no repeated factor, $\bar{A}$ and $\bar{C}$ have no common factor. If $\bar{A}$ divides $\bar{C}h$, it therefore divides h , and then $h = 0$ because $\deg h < d$. The linear map (3.16) is thus injective, hence invertible.

Step 3: integrality and rigidity. A solution could have ℓ in a denominator, and then it could not be reduced. We show that this does not happen. The argument is a dominant balance, with ℓ in the role of the small parameter: a solution that is large is rescaled to size one, and the leading-order problem, which is the continuous one, then leads to a contradiction. For $\lambda \neq 0$ and a monic polynomial A of degree d , put $A_\lambda(\rho) = \lambda^{-d}A(\lambda\rho)$. This is the monic polynomial whose roots are those of A divided by λ , and its coefficients are $\alpha'_j = \alpha_j/\lambda^j$. A polynomial T in $\alpha_1, \dots, \alpha_d$ and ρ that is weighted-homogeneous of weight w satisfies $T(\alpha; \lambda\rho) = \lambda^w T(\alpha'; \rho)$.

Let A be a solution of the matching equations with coefficients in K , and suppose that some root of A is not ℓ -integral. Let λ be a root of least valuation, so that $\theta = \text{ord}(\lambda) < 0$, and put $B = A_\lambda$. The roots of B are ℓ -integral, hence so are its coefficients α'_j , by (V). Decompose $\ell S_A = \sum_{w \leq \ell} T^{(w)}$ into its parts of weight w . They have coefficients in $\mathbb{Z}_{(\ell)}$ by Step 1, and $T^{(\ell)} = \ell H_A$. Since A divides S_A and $A(\lambda) = 0$, we have $S_A(\lambda) = 0$. Dividing this equation by λ^ℓ gives

$$\ell H_B(1) = - \sum_{w < \ell} \lambda^{w-\ell} T^{(w)}(\alpha'; 1).$$

Each term on the right has positive valuation, because $\text{ord}(\lambda^{w-\ell}) = (w-\ell)\theta > 0$ and $T^{(w)}(\alpha'; 1)$ is ℓ -integral. So the right side reduces to zero. The left side reduces to $(-1)^d \neq 0$, by (3.15) evaluated at $\alpha = \alpha'$ and $\rho = 1$. This is a contradiction. Hence every root of A is ℓ -integral, and so is every coefficient. The solution can be reduced, and by Step 2 its reduction is one of the polynomials (3.9).

Rigidity is proved in the same way. Suppose that $A \neq \rho^d$ is monic of degree d , with complex coefficients, and divides H_A . Since $H_{A_\lambda}(\rho) = \lambda^{-\ell} H_A(\lambda\rho)$, every A_λ has the same property, so we may assume that one of the coefficients α_j equals 1. By (N), applied to the equations $\text{rem}_A H_A = 0$ together with this normalization, we may assume that the coefficients of A lie in $\mathbb{Q} \subset K$. Let λ be a root of A of least valuation; it is not zero, because $A \neq \rho^d$. Put $B = A_\lambda$. As before, B has ℓ -integral coefficients. Since A divides H_A and $A(\lambda) = 0$, we have $0 = \ell H_A(\lambda) = \lambda^\ell \ell H_B(1)$. But $\ell H_B(1)$ reduces to $(-1)^d \neq 0$, by (3.15). This contradiction proves (c).

Step 4: lifting. The system $(\ell R_r)_{0 \leq r < d}$ has coefficients in $\mathbb{Z}_{(\ell)}$, and by Step 2 each of the $\binom{\ell}{d}$ reduced solutions lies in $\mathbb{F}_\ell^d$ and is simple. By Hensel's lemma (H), each of them is the reduction of exactly one solution in $\mathcal{O}^d$. By Step 3 every solution in K^d lies in $\mathcal{O}^d$, and its reduction is one of the reduced solutions. Hence reduction modulo ℓ is a bijection between the solutions in K^d and the reduced solutions. There are exactly $\binom{\ell}{d}$ solutions in K^d , and the Jacobian determinant is nonzero at each of them, because its reduction is nonzero. By (T), the matching equations have the same $\binom{\ell}{d}$ solutions over $\mathbb{C}$, with coordinates in $\overline{\mathbb{Q}}$, all simple. This proves (a) and (b); part (c) was proved in Step 3. $\square$

Remark 3.5 (Formal verification). Parts (a) and (c) of Theorem 3.4 have been verified in Lean 4 against the Mathlib library, both for the matching equations and for the pairs (P, v) themselves (Appendix A). The formal proof follows the route above, and like the proof above it normalizes the roots of A .

3.4 Reading the labels: pulses and fronts

As $z \rightarrow +\infty$ we have $Q \rightarrow 1$ and $D^j Q \rightarrow 0$ for $j \geq 1$, so $F(D)Q \rightarrow F(0)$ for every polynomial F . The endpoints of v_A , its limits as $z \rightarrow \mp\infty$, are therefore 0 and $s_p A(0)$. Pairs with $A(0) = 0$ have equal endpoints and are pulses; the others are fronts. The pulses are needed separately, because the elliptic pairs of Section 4 degenerate onto them.

Letting $z \rightarrow +\infty$ in (3.4) gives $S_A(0) = -A(0)^2$. Since $S_A(0) = A(0)C_A(0) + R_0$ by (3.5), the constant remainder satisfies the polynomial identity

$$R_0 = -A(0)(A(0) + C_A(0)). \quad (3.17)$$

On the hyperplane $\alpha_d = A(0) = 0$ the equation $R_0 = 0$ therefore holds identically, and the pulses are the solutions of the *pulse equations*

$$R_1 = \cdots = R_{d-1} = 0 \quad \text{in the unknowns } \alpha_1, \dots, \alpha_{d-1}, \quad \alpha_d = 0. \quad (3.18)$$

We write $\mathcal{Z}_p$ for the set of their solutions; the multiplicity of a point of $\mathcal{Z}_p$ is its multiplicity as a solution of the pulse equations.

Proposition 3.6 (Pulses and fronts). *Let $\ell = 2p - 1$ be prime. A pair is a pulse if and only if its label contains 0. Consequently exactly $S_p := \binom{2p-2}{p-2}$ of the N_p pairs are pulses, and the remaining $\binom{2p-2}{p-1}$ are fronts. Every pulse is a simple solution of the pulse equations (3.18), and $P_A(0) \neq 0$ at every pulse (these two facts are used in Section 4). These are counts over $\mathbb{C}$.*

In Table 2 the pulses are the four pairs with $b = 0$, and their labels are the four that contain 0.

Proof. Let A be a solution with label L . If $A(0) = 0$, then $\overline{A}(0) = 0$, so $0 \in L$. Conversely, let $0 \in L$. At the solution A we have $\ell S_A = A \cdot \ell C_A$. By (V) and (3.14), ℓC_A is ℓ -integral and reduces to $(-1)^d \overline{C}$, where $\overline{C} = (\rho^\ell - \rho)/\overline{A}$. Since ρ divides $\overline{A}$ and $\rho^\ell - \rho$ has no repeated factor, $\overline{C}(0) \neq 0$. The number $\ell A(0)$ reduces to zero. Hence $\ell(A(0) + C_A(0))$ reduces to $(-1)^d \overline{C}(0) \neq 0$. So $A(0) + C_A(0) \neq 0$, and $R_0 = 0$ gives $A(0) = 0$, by (3.17): the pair is a pulse. The number of d -element subsets of $\mathbb{F}_\ell$ that contain 0 is $\binom{\ell-1}{d-1} = S_p$.

At a pulse, (3.17) shows that the row of the Jacobian matrix of the matching equations that belongs to R_0 has a single entry that can be different from zero, namely $-C_A(0)$ in the

column of α_d . The Jacobian matrix is invertible, by Theorem 3.4(a). Hence $C_A(0) \neq 0$, so that $P_A(0) = s_p C_A(0)/2 \neq 0$, and the minor obtained by deleting that row and that column is invertible. This minor is the Jacobian matrix of the pulse equations. (For $d = 1$ there are no pulse equations, and only $C_A(0) \neq 0$ is asserted.) $\square$

Mirror images. Reading v_A in the opposite direction, and restoring the normalization, gives again a pair, with $A(\rho)$ replaced by $(-1)^d A(-\rho)$ (Section 6). The roots of $\bar{A}$ change sign, so the label L becomes $-L$. In Table 2 this is $(a, b) \mapsto (-a, b)$.

4 The count of elliptic pairs

We now count the elliptic pairs of Section 2 on a fixed lattice $\Lambda \subset \mathbb{C}$, and put $g_i = g_i(\Lambda)$. The pole is at zero, and the constant term of the profile is retained. Neither reflection nor a lattice automorphism is factored out: pairs related in this way are counted separately (so that $M_3 = 8$ below, and not fewer). Two lattices differ by a rescaling exactly when they have the same invariant $j(\Lambda) = 1728g_2^3/(g_2^3 - 27g_3^2)$.

We write elliptic functions through the Weierstrass relations

$$Y^2 = 4X^3 - g_2X - g_3, \quad \mathcal{D}X = Y, \quad \mathcal{D}Y = 6X^2 - \frac{g_2}{2}. \quad (4.1)$$

For $g_2^3 \neq 27g_3^2$, that is, on a lattice, these relations are realized by $X = \wp(z; g_2, g_3)$, $Y = \wp'(z; g_2, g_3)$ and $\mathcal{D} = d/dz$; see NIST [69, Chapter 23]. We use X and Y because the formulas below must keep their meaning for the degenerate parameters $g_2^3 = 27g_3^2$, which belong to no lattice. Using (4.1), every polynomial in X and Y reduces to a unique normal form $F(X) + YG(X)$. We say that $X^k Y^\epsilon$, $\epsilon \in \{0, 1\}$, has pole order $2k + 3\epsilon$; on a lattice this is its pole order at zero.

The representation of the profile is the classical one. An elliptic function with one pole per period has zero residue there. The functions $1, \wp, \wp', \dots, \wp^{(p-2)}$ have distinct principal parts and span the space of elliptic functions whose only pole is at zero, of order at most p . Subtracting the indicated principal part from a one-pole solution leaves an entire elliptic function, hence a constant. Since $D^{p-2}\wp = (-1)^p(p-1)!z^{-p} + O(1)$, the leading coefficient (2.7) shows that every profile has the form

$$V = -s_p \mathcal{D}^{p-2}X + \sum_{j=0}^{p-3} u_j \mathcal{D}^j X + b. \quad (4.2)$$

Its only pole on $\mathbb{C}/\Lambda$ is zero, of order p . In particular Λ is its full period lattice. The matching equations $\mathcal{E}_p(g_2, g_3)$ are obtained by formal substitution of (4.2) in (1.1), using only (4.1): the left side is reduced to its normal form, and the coefficients of F and G are set equal to zero. The unknowns are $a_0, \dots, a_{p-1}$, $u_0, \dots, u_{p-3}$ and b . The equations are polynomials in the unknowns and in g_2, g_3 , and they are defined for every $(g_2, g_3) \in \mathbb{C}^2$, the degenerate parameters included; on a lattice their solutions are the elliptic pairs. The symbol $\mathcal{E}_p(g_2, g_3)$ also denotes the set of solutions. Multiplicity and simplicity are understood as in Section 3.

Theorem 4.1 (Fixed-lattice elliptic enumeration). *Let $2p - 1$ be prime.*

- (a) *On every lattice the elliptic pairs are finite in number, and counted with multiplicity there are*

$$M_p := 2 \binom{2p-2}{p-2} = 2S_p \quad (4.3)$$

of them.

- (b) *There is a proper algebraic subset $\mathcal{N}$ of the (g_2, g_3) -plane outside which all of them are simple. The lattices in $\mathcal{N}$ are those whose invariant $j(\Lambda)$ lies in a finite set, which depends on p . Thus a generic lattice, that is, a lattice whose invariant avoids these values, admits exactly M_p distinct elliptic pairs.*

The smallest case. For $p = 2$ the profile is $V = -12X + b$, and the left side of (1.1) reduces to

$$-12a_1Y - 12(a_0 + b)X + 6g_2 + a_0b + \frac{1}{2}b^2.$$

Consequently the complete elliptic system is

$$a_1 = 0, \quad v = -12\wp - a_0, \quad a_0^2 = 12g_2. \quad (4.4)$$

There is no one-pole elliptic member with nonzero Burgers coefficient: both elliptic pairs belong to the KdV equation, in agreement with $M_2 = 2$. They are the two backgrounds (2.5) of one cnoidal wave. For $g_2 \neq 0$ they are distinct and simple. For $g_2 = 0$, $g_3 \neq 0$, they coincide, and the lattice carries one pair of multiplicity two. Here $\mathcal{N}$ is the line $g_2 = 0$, and the finite set of (b) is $\{0\}$. At order three it is $\{0, -300\}$ (Section 8.2).

No escape to infinity. The number M_p is twice the number S_p of pulses of Proposition 3.6, and the proof shows why. When the parameters of a square polynomial system (as many equations as unknowns) vary, its solutions may merge or separate. As long as they remain finite in number, their number, counted with multiplicity, can change only if a solution moves off to infinity. A solution that moves off to infinity, rescaled to size one, converges to a nonzero solution of the leading parts of the equations; this is the rescaling argument of Step 3 in the proof of Theorem 3.4, with the ordinary absolute value in place of the ℓ -adic one. Here the leading parts are the equations with both periods infinite ($\wp \rightarrow z^{-2}$; the cusp $g_2 = g_3 = 0$), and by rigidity that rational problem has only the trivial solution. So nothing escapes, and the number of pairs, counted with multiplicity, is the same for all (g_2, g_3) . We find it with one period infinite ($\wp$ becomes a hyperbolic function; the node): there every elliptic pair becomes a pulse on one of its two constant backgrounds, and every pulse on either background arises in this way exactly once. The pulses are simple, and simplicity spreads from the node to a generic lattice.

Lemma 4.2 (Conservation of number). *Let $E_1, \dots, E_n$ be polynomials in unknowns $x = (x_1, \dots, x_n)$ and parameters $t = (t_1, \dots, t_m)$, weighted-homogeneous for some positive integer weights of the x_j and the t_k . Suppose that the leading system $E_i(x; 0) = 0$, $1 \leq i \leq n$, has no solution in $\mathbb{C}^n$ other than $x = 0$. Then:*

- (i) *for every t the system $E(x; t) = 0$ has finitely many solutions, and they stay bounded while t stays bounded;*
- (ii) *the number of solutions, counted with multiplicity, is the same for every $t \in \mathbb{C}^m$;*
- (iii) *the t for which some solution is not simple form an algebraic subset of $\mathbb{C}^m$, that is, the common zero set of finitely many polynomials in t .*

The proof of the lemma uses more algebra than the rest of the paper and can be skipped; Steps 1–5 below use only its statement.

Proof. Since the leading system has only the zero solution, the polynomials in x modulo its equations form a finite-dimensional space [11, Chapter 5, §3], spanned by monomials $m_1 = 1, \dots, m_r$. Every polynomial f in (x, t) is then, modulo the equations E_i , a combination $\sum_i c_i(t)m_i$ with polynomials c_i in t . This is proved by induction on the weight of f , which may be taken weighted-homogeneous: write $f = f(x, 0) + \sum_k t_k f_k$, write $f(x, 0)$ as a combination of the m_i plus $\sum_i h_i E_i(x; 0)$ with weighted-homogeneous h_i , and replace each $E_i(x; 0)$ by E_i minus its terms that contain some t_k . What remains is a sum of products of a t_k with a polynomial of lower weight, because t_k has positive weight. In particular $x_j m_i = \sum_l M_{il}^{(j)}(t)m_l$ modulo the equations. Multiplying this vector identity by the adjugate of the matrix $x_j - M^{(j)}(t)$ shows that $\det(x_j - M^{(j)}(t)) m_i$ is a combination of the E_i for every i (the Cayley–Hamilton argument); for $m_1 = 1$ this is the determinant itself. So at every solution x_j is a root of a monic polynomial of degree r whose coefficients are polynomials in t . This proves (i).

For (ii), join two parameter values by a segment. By (i) some ball $|x| < \varrho$ contains all solutions for all t on the segment, so no solution crosses the sphere $|x| = \varrho$. The number of zeros of a holomorphic map in a ball, counted with multiplicity, when they are finite in number and none lies on the bounding sphere, is the degree of $E/|E|$ on that sphere, and it does not change under a deformation without zeros on the sphere [64, Appendix B], [2, Sections 5.2 and 5.4]; this is Rouché’s theorem in several variables.

For (iii), the set in question is the projection to $\mathbb{C}^m$ of the algebraic set on which $E_1, \dots, E_n$ and the Jacobian determinant $\det(\partial E_i / \partial x_j)$ vanish. The monic polynomials of (i) are combinations of the E_i . Eliminate the unknowns one at a time: at each step the polynomial of (i) for the unknown being eliminated has leading coefficient 1, so by the Extension Theorem of elimination theory [11, Chapter 3, §1] every solution of the eliminated equations extends. The projection is therefore the common zero set of the polynomials in t alone that are combinations of the E_i and the Jacobian determinant. $\square$

Three operations do not change multiplicities, and the proof of the theorem uses them freely: an invertible affine change of the unknowns; invertible linear combinations of the equations with constant coefficients; and the elimination of an unknown that one of the equations gives as a polynomial in the others. Each of them leaves unchanged, up to an isomorphism, the space of power series modulo the combinations of the equations, whose dimension is the multiplicity (Section 3).

Proof of Theorem 4.1. Step 1: eliminating the operator. The coefficient at pole order $2p$ vanishes by the choice of s_p . The coefficients at orders $2p-1, \dots, p$ determine $a_{p-1}, \dots, a_0$ successively. At each step the coefficient of the new a_j is the nonzero rational leading coefficient of $D^j V$. This is a polynomial elimination: each a_j is obtained as a polynomial in the remaining unknowns and in g_2, g_3 , and multiplicities do not change. The residual has pole order at most $p-1$ and zero residue, since no normal form has pole order one. It lies in the span of 1 and the $\mathcal{D}^j X$, $0 \leq j \leq p-3$. Hence $p-1$ equations $E_1 = \dots = E_{p-1} = 0$ remain in the $p-1$ profile coefficients u_j, b : the coefficients of the residual in this basis.

Step 2: weights. Give D, X, g_2, g_3 weights 1, 2, 4, 6, respectively, and give the profile weight p ; this is the usual homogeneity of $\wp$ under a rescaling of z . Then u_j has weight $p-j-2$, and b has weight p . After the triangular elimination, the residual principal-part coefficients and the residual constant, that is, the E_i , are weighted-homogeneous polynomials in the u_j, b, g_2, g_3 , of weights

$$\begin{array}{l|l} \text{variables} & 1, 2, \dots, p-2, p \\ \text{equations} & p+1, p+2, \dots, 2p-2, 2p. \end{array} \quad (4.5)$$

The Bézout number of this table is $(p+1) \cdots (2p-2)(2p)/((p-2)!p) = 2\binom{2p-2}{p-2} = M_p$ (at $p=3$: $4 \cdot 6/(1 \cdot 3) = 8$), in agreement with the count below (companion note). The normal forms and the eliminations are polynomial in g_2, g_3 , so these statements extend to the degenerate parameters. For fixed g_2, g_3 , the highest weighted parts of the equations in the profile variables are obtained by setting $g_2 = g_3 = 0$, since g_2 and g_3 carry positive weight.

Step 3: the cusp. At the cusp $g_2 = g_3 = 0$ the relations (4.1) are satisfied by $X = z^{-2}$, $Y = -2z^{-3}$ and $\mathcal{D} = d/dz$, so a solution of the cusp system gives a nonconstant rational solution V of (1.1) whose only pole is at zero. By Lemma 3.3, V vanishes at infinity, so that $b = 0$; moreover $a_0 = P(0) = 0$, and $V = (-1)^{p-1} s_p A^\sharp$ for a monic A of degree $p-1$ that divides H_A . By rigidity, which holds by Theorem 3.4(c), $A = \rho^{p-1}$. Thus $V = c_* z^{-p}$, and all the variables in (4.2) vanish, because the functions $D^j z^{-2}$ have different pole orders. The leading equations have no nonzero common zero.

Step 4: conservation of number. The residual equations $E_1, \dots, E_{p-1}$ of Step 1 are polynomials in the $p-1$ profile unknowns and in g_2, g_3 , weighted-homogeneous for the weights of Step 2, and by Step 3 their leading system, which is the cusp system, has only the zero solution. Lemma 4.2 applies, with $t = (g_2, g_3)$. By (i), every system $\mathcal{E}_p(g_2, g_3)$ has finitely many solutions, which proves the finiteness in (a). By (ii), their number, counted with multiplicity, is the same at every point of the (g_2, g_3) -plane, the degenerate parameters included; at the cusp all of this multiplicity sits at the single solution where every unknown vanishes. By (iii), the parameters at which some solution is not simple form an algebraic subset $\mathcal{N}$ of the plane.

Step 5: the node. To find the common number, and to show that $\mathcal{N}$ is not the whole plane, consider the nodal parameters

$$g_2 = \frac{1}{12}, \quad g_3 = -\frac{1}{216}. \quad (4.6)$$

For these parameters the relations (4.1) are satisfied by

$$X = Q^2 - Q + \frac{1}{12}, \quad Y = (2Q-1)Q(1-Q), \quad \mathcal{D} = D \quad \text{with} \quad DQ = Q(1-Q);$$

this is the degenerate $\wp$ -function with the single period $2\pi i$, with its pole moved to $z = i\pi$, written in the logistic coordinate. A monomial $X^k Y^\epsilon$ becomes a polynomial in Q of degree $2k+3\epsilon$, which is also its pole order. These degrees are all different, so the monomials remain linearly independent, and the matching equations for the nodal parameters are exactly the conditions for (1.1) to hold as an identity between polynomials in Q . Since $X = \frac{1}{12} - DQ$ and $\mathcal{D}^j X = -D^{j+1}Q$ for $j \geq 1$, the profile (4.2) becomes, with $u_{p-2} = -s_p$,

$$V = \beta + v_A, \quad A(\rho) = -\frac{1}{s_p} \sum_{j=0}^{p-2} u_j \rho^{j+1}, \quad \beta = b + \frac{u_0}{12}.$$

Here A is monic of degree $d = p-1$ with $A(0) = 0$, and β is the common limit of V at the two ends. With the coefficients α_j of A as in (3.1), the passage from $(u_0, \dots, u_{p-3}, b)$ to $(\alpha_1, \dots, \alpha_{d-1}, \beta)$ is an invertible affine change of unknowns with rational coefficients.

In these unknowns the constant term of $P(D)V + V^2/2$ is $P(0)\beta + \beta^2/2$, and the remaining terms are $(P+\beta)(D)v_A + v_A^2/2$. By the proof of Proposition 3.1 the latter equals $(s_p/2)[2(P+\beta)A - s_p S_A](D)Q$, and the elimination of Step 1 is the division of S_A by A : it gives $P+\beta = s_p C_A/2$, a polynomial in the unknowns, which we still denote by P_A , and it leaves $-(s_p^2/2)R_A(D)Q$. Since $A(0) = 0$, the coefficient R_0 vanishes identically, by (3.17). With $P(0) = P_A(0) - \beta$, the nodal matching equations are therefore invertible linear combinations, with constant coefficients, of

$$R_1 = \dots = R_{d-1} = 0, \quad \beta(\beta - 2P_A(0)) = 0, \quad (4.7)$$

together with $P = P_A - \beta$. These are the pulse equations (3.18) and one quadratic equation for the background, and the two systems have the same solutions with the same multiplicities.

By Proposition 3.6 the pulse equations have exactly S_p solutions, all simple, with $P_A(0) \neq 0$ at each of them. For each of them the quadratic has the two simple roots $\beta = 0$ and $\beta = 2P_A(0)$, and the Jacobian matrix of (4.7) is block triangular. For the nodal parameters the system thus has exactly $2S_p$ solutions, all simple. By Step 4 the number of solutions counted with multiplicity is $2S_p = M_p$ for all (g_2, g_3) . This proves (a). Moreover the nodal parameters do not belong to $\mathcal{N}$. So $\mathcal{N}$ is a proper algebraic subset of the plane, and outside $\mathcal{N}$ every solution is simple. A nonzero polynomial cannot vanish on a nonempty open subset of the plane, so $\mathcal{N}$ does not contain all the lattices. It contains only those with finitely many values of j , as we now show.

The equations E_i and their Jacobian determinant are weighted-homogeneous, so $\mathcal{N}$ is invariant under $(g_2, g_3) \mapsto (\lambda^4 g_2, \lambda^6 g_3)$, $\lambda \neq 0$. On lattices this is the rescaling $\Lambda \mapsto \lambda^{-1} \Lambda$, which maps pairs to pairs and simple pairs to simple pairs. If a polynomial vanishes on $\mathcal{N}$, then so does each of its weighted-homogeneous components. Since $\mathcal{N}$ is a proper algebraic subset, it therefore lies in the zero set of a nonzero polynomial that is a monomial in g_2, g_3 times a product $\prod_i (g_2^3 - c_i g_3^2)$. On the lattices this zero set consists of finitely many values of j : $j = 0$ on $g_2 = 0$, $j = 1728$ on $g_3 = 0$, and $j = 1728c_i/(c_i - 27)$ on $g_2^3 = c_i g_3^2$. Two lattices with the same j differ by a rescaling, so $\mathcal{N}$ contains both or neither. Hence the lattices in $\mathcal{N}$ are exactly those whose invariant $j(\Lambda)$ lies in a finite set, which depends on p . This proves (b). $\square$

Two backgrounds. The factor two in M_p retains the two choices of background in (2.5). That transformation preserves the evolution equation and the lattice; it changes the operator P and the speed. For the nodal parameters it exchanges the two roots of the quadratic in (4.7). Outside $\mathcal{N}$ the two backgrounds of every wave are distinct, because a solution with $a_0 = 0$ is never simple: Step 1 gives $a_0 + b$ as a polynomial in the u_j and g_2, g_3 , and b enters the residual equations only through the constant one, which gives a_0^2 as another such polynomial; so the Jacobian determinant contains the factor a_0 . On a generic lattice the M_p pairs therefore represent $M_p/2$ elliptic waves, up to translations and Galilean shifts, each counted with both of its backgrounds. On special lattices the two backgrounds may coalesce, as (4.4) shows for $g_2 = 0$ and Section 8.2 shows at order three.

The elliptic pairs of purely dispersive equations, $2F_p$ of the M_p , are counted in Section 6 (Corollary 6.4), after the symmetry of their profiles has been established; at order two they are both pairs of (4.4), and $M_2 = 2F_2 = 2$.

5 Are these all the meromorphic waves? The classification theorem

Sections 3 and 4 counted the one-pole waves that are rational in an exponential or elliptic, for every operator P . They did not ask whether an equation can have other meromorphic travelling waves. For two classes of equations it cannot (Theorem 5.1). The proof is a local analysis at a pole of the kind used in the Painlevé test, Laurent series and resonances, combined with Eremenko's theorem. A by-product, Corollary 5.3, is what identifies the two pure subfamilies in Section 6.

Theorem 5.1 (Classification). *Assume either that p is odd or that P is an even polynomial; that is, the evolution equation has even order or is purely dispersive. Every nonconstant meromorphic solution of (1.1) is of one of the three kinds of Section 2.3: rational in z , rational in $e^{\kappa z}$ for some*

$\kappa \neq 0$, or elliptic. Every pole has order p , and the poles are the translates of one pole by the full group of periods: the solution has one pole per period, in the sense of Section 2.3.

For fixed P of odd degree, there is at most one nonconstant solution with its pole at zero. For even P , the same statement holds after fixing the value of the first integral (5.5) below, the energy. The only entire solutions are $v = 0$ and $v = -2a_0$.

Remark 5.2 (The hypothesis cannot be dropped: the Fisher family). For even p the theorem requires a purely dispersive equation, and this cannot simply be dropped. The known example comes from the Fisher equation. Its travelling waves of nonzero speed satisfy (1.1) with $p = 2$ and a mixed operator, which is also the profile equation of KdV–Burgers. At the special speed of Ablowitz and Zeppetella [1] it has a one-parameter family of solutions, given in Kudryashov [37], Conte et al. [7, Lemma 4.4] and McCue et al. [63], and for the KdV–Burgers equation itself in Kudryashov [46]; see also Kudryashov [35]. In our normalization it is

$$v(z) = -12\kappa^2 e^{2\kappa z} \wp(e^{\kappa z} - 1; 0, g_3), \quad P(D) = D^2 - 5\kappa D + 6\kappa^2, \quad \kappa \neq 0.$$

Substitution follows from $(D^2 - 5\kappa D + 6\kappa^2)[t^2 U(t)] = \kappa^2 t^4 U''(t)$, $t = e^{\kappa z}$. For $g_3 = 0$ and $\kappa = 1$ this is the front in the first row of (8.2): the familiar Ablowitz–Zeppetella front is a rational-exponential degeneration of the family. For generic $g_3 \neq 0$ the solution is globally meromorphic and is of none of the three kinds. Further examples of meromorphic solutions of none of the three kinds are discussed by Ng and Wu [67, Remark 1.11]. Thus a classification for mixed operators with p even must allow a larger class of functions. For mixed operators of even degree $p \geq 4$ we do not know whether meromorphic solutions of other kinds occur. Methods for constructing elliptic and rational-exponential solutions without the finiteness property are developed by Demina [17].

Earlier work. Theorem 5.1 is a uniform version of results that exist in pieces; for the general analytic background, see Eremenko [21]. The case of odd p is contained in Yuan et al. [74, Theorem 1], with related results for odd p in Huang et al. [25], Yuan et al. [75] and a fifth-order instance in Dang [13]. For even P , fixing the value of a first integral to remove the freedom at the resonance is the device of Eremenko et al. [23, Lemmas 3–4] for $P = D^p + a_0$, stated as a general step by Conte et al. [8, Section 3.2] and Conte et al. [9, Section 5] and applied to the Kawahara equation [18, 73], the generalized Bretherton equation [19, 60] and the seventh-order KdV equation [12]; the familiar Kawahara first integral is the case $m = 2$, that is $p = 4$, of (5.5) below. Theorem 5.1 extends it uniformly to every even P , intermediate derivatives included. Equations with two dominant monomials, or without the finiteness property (finitely many Laurent expansions at a pole), are treated by other methods in Demina [15, 17].

5.1 The Laurent series at a pole

We now prove Theorem 5.1. Write the expansion at a pole at zero as

$$v(z) = \sum_{n \geq 0} c_n z^{n-p}. \quad (5.1)$$

The pole order p and the leading coefficient $c_0 = c_*$ were found in (2.7), for every P . For the remaining coefficients, set

$$I_p(n) = \prod_{j=p}^{2p-1} (n-j) + c_*. \quad (5.2)$$

The product is the factor produced by D^p acting on z^{n-p} , and c_* is the factor produced by the cross term of $c_n z^{n-p}$ with the leading pole in $v^2/2$. By (2.7), c_* has the sign $(-1)^{p-1}$, and $|c_*| = 2p(p+1) \cdots (2p-1)$ is twice the absolute value of the product at $n = 0$. With $c_n = 0$ for $n < 0$, comparison of the coefficients of z^{n-2p} in (1.1) gives

$$I_p(n)c_n = - \sum_{j=0}^{p-1} a_j \prod_{i=1}^j (n-2p+i) c_{n-p+j} - \frac{1}{2} \sum_{k=1}^{n-1} c_k c_{n-k}, \quad n \geq 1. \quad (5.3)$$

The right side contains only coefficients c_k with $k < n$, so (5.3) determines c_n whenever $I_p(n) \neq 0$. The positive integer zeros of I_p are the *resonances* (Fuchs indices) of the recurrence: at a resonance n the recurrence leaves c_n undetermined if its right side vanishes, and has no solution otherwise.

If p is odd, then $c_* > 0$, and there is no positive integer resonance. For integers $1 \leq n < p$, the product in (5.2) is negative with absolute value less than $c_*/2$. For $p \leq n \leq 2p-1$ it is zero, and for $n \geq 2p$ it is positive. Hence $I_p(n) \neq 0$ for every positive integer n , and there is exactly one formal Laurent expansion at the pole.

Theorem 2 of Eremenko [22] states that an autonomous polynomial differential equation with finitely many possible formal pole expansions and a single monomial of highest total degree has all its meromorphic solutions among the three kinds. In (1.1) the only monomial of degree two is $v^2/2$, and for odd p we have just seen that there is exactly one formal pole expansion. Both hypotheses hold at every odd order, for all coefficients a_j .

5.2 The energy argument for a purely dispersive equation

Let $p = 2m$ and $P(D) = D^{2m} + \sum_{j=0}^{m-1} a_{2j} D^{2j}$. Now $c_* < 0$, and there is exactly one positive integer resonance. Indeed the product in (5.2) is at most $|c_*|/2$ for $0 \leq n < p$, is zero for $p \leq n \leq 2p-1$, and is strictly increasing for $n \geq 2p$; its value at $3p$ is $(p+1)(p+2) \cdots (2p) = -c_*$. The only positive integer zero of I_p is therefore $n = 3p$, and the possible free Laurent coefficient is c_{3p} , the coefficient of z^{2p} in v . If the recurrence is compatible at $n = 3p$, the equation has a one-parameter family of formal expansions at the pole, and Eremenko's theorem does not apply to (1.1) itself. The remedy is a first integral whose value determines c_{3p} .

For $j \geq 1$, define the quadratic differential expression

$$\mathcal{B}_j(v) = \sum_{i=0}^{j-2} (-1)^i v^{(i+1)} v^{(2j-1-i)} + \frac{(-1)^{j-1}}{2} (v^{(j)})^2, \quad (5.4)$$

where the sum is empty when $j = 1$. Termwise differentiation gives $D\mathcal{B}_j = v'v^{(2j)}$: every even derivative, multiplied by v' , is an exact derivative. Since P contains only even derivatives, it follows that

$$\mathcal{H}_P(v) = \mathcal{B}_m(v) + \sum_{j=1}^{m-1} a_{2j} \mathcal{B}_j(v) + \frac{a_0}{2} v^2 + \frac{1}{6} v^3 \quad (5.5)$$

satisfies

$$D\mathcal{H}_P(v) = v' \left(P(D)v + \frac{1}{2} v^2 \right). \quad (5.6)$$

In particular $\mathcal{H}_P$, the energy, is constant on each solution. It is the energy in the usual sense: an even operator $P(D)$ is formally self-adjoint, (1.1) is then the Euler–Lagrange equation of an autonomous Lagrangian, and $\mathcal{H}_P$ is the conserved quantity that belongs to translation in z .

The energy identity has two consequences for the Laurent expansion. The first is that the resonance is always compatible. Construct (5.1) through c_{3p-1} ; the first possible residual of (1.1) is Ωz^p , and multiplication by v' gives residue $-pc_*\Omega$. By (5.6) this is the residue of a derivative of a Laurent series, and is therefore zero. Hence $\Omega = 0$, and c_{3p} may be chosen arbitrarily.

The second consequence is that the energy determines c_{3p} . On the full formal solution the constant energy has the form

$$\mathcal{H}_P = -pc_*I'_p(3p)c_{3p} + (\text{a constant that depends only on } P), \quad (5.7)$$

and the coefficient of c_{3p} is not zero: the product in (5.2) equals $-c_*$ at $n = 3p$, so that $I'_p(3p) = -c_* \sum_{j=p+1}^{2p} 1/j > 0$. To find the coefficient of c_{3p} , give z the weight -1 and D the weight 1 , so that v has weight p . The highest-weight part of (5.6) is then the part without the a_{2j} . Linearize it about c_*z^{-p} , with perturbation z^{n-p} . If the corresponding energy term is $\omega(n)z^{n-3p}$, then $(n-3p)\omega(n) = -pc_*I_p(n)$, and $n \rightarrow 3p$ gives the coefficient in (5.7). Lower derivative terms have different powers of z and cannot contribute to this coefficient. Fixing the energy thus fixes c_{3p} , and all subsequent Laurent coefficients follow from (5.3). This is the resonance-removal step of the works cited above, with an explicit coefficient for the present operator family.

We can now apply Eremenko's theorem, not to (1.1) but to the level equation, in which P and the value of $\mathcal{H}_P(v)$ are fixed. This equation is autonomous and polynomial, and its unique monomial of highest degree is $v^3/6$. It has exactly one formal pole expansion: differentiating it and dividing by the nonzero Laurent series v' returns (1.1), so every expansion belongs to the family above, and (5.7) selects the value of c_{3p} . Every meromorphic solution of (1.1) lies on one energy level and is therefore of one of the three kinds.

5.3 The period group and the entire solutions

It remains to show that the poles of one solution are the translates of one pole by its periods. In either case of the theorem, take two poles z_1, z_2 of the same solution. The functions $v(z+z_1)$ and $v(z+z_2)$ satisfy the same equation and, in the even case, have the same energy. Their Laurent series at zero coincide, so the identity theorem gives $v(z+z_1) = v(z+z_2)$. Every difference of poles is a period. Conversely, every period translates a pole to a pole. The period group is discrete: a sequence of nonzero periods tending to zero would force $v' \equiv 0$. The same argument proves the uniqueness statements of the theorem: two nonconstant solutions with the same operator, the same energy in the even case, and a pole at zero have the same Laurent series there.

Finally, an entire solution of one of the three kinds must be constant. In the rational case it is a polynomial in z , and in the rational-exponential case a Laurent polynomial in $e^{\kappa z}$. In both cases squaring doubles an extreme nonzero exponent, which $P(D)v$ cannot cancel (Section 2.3). An entire elliptic function is constant. The constants solve $a_0v + v^2/2 = 0$, which gives $v = 0$ and $v = -2a_0$. This completes the proof of Theorem 5.1.

The same uniqueness shows that, in the two pure subfamilies, every wave inherits the reflection symmetry of its equation.

Corollary 5.3 (Waves are symmetric about their poles). *Let v be a nonconstant meromorphic solution of (1.1) and let z_0 be one of its poles.*

- (i) *If P is even, then $v(z_0 + \zeta) = v(z_0 - \zeta)$.*
- (ii) *If p is odd and $P - P(0)$ is odd, then $v(z_0 + \zeta) + P(0) = -(v(z_0 - \zeta) + P(0))$.*

Consequently a solution that is rational in an exponential is a pulse in case (i). In case (ii) its two limits add up to $-2P(0)$, so it is a front unless $P(0) = 0$.

For example, the KdV soliton $3\text{sech}^2(z/2)$ is even about its pole $i\pi$, and the Kuramoto–Tsuzuki front, the row $(a, b) = (0, 1/11)$ of Table 2, has $v + P(0) = \frac{15}{11}(11\tau^3 - 9\tau)$ with $\tau = \tanh(z/2)$, which is odd about $i\pi$.

Proof. In case (i) the function $v(z_0 - \zeta)$ solves the same equation, has a pole at $\zeta = 0$, and has the same energy as v , because every term of (5.4) and (5.5) contains an even total number of derivatives. Fixed-energy uniqueness gives (i). In case (ii) the function $-v(z_0 - \zeta) - 2P(0)$ solves the same equation, by Section 2, and has a pole at $\zeta = 0$. For odd p the Laurent expansion at a pole is unique, which gives (ii). The two limits exist by Section 2.3, and the statements about them follow by letting ζ tend to the two ends. $\square$

6 Mirror images and the two pure subfamilies

The numbers N_p and M_p refer to the full monic family. We now count the pairs of the two pure subfamilies. These are the pairs that coincide with their own mirror image under the reflection $z \mapsto -z$. On the labels the reflection is $L \mapsto -L$, and the number of labels with $L = -L$ is again a binomial coefficient.

For a monic polynomial A of degree $d = p - 1$ put

$$(\iota A)(\rho) = (-1)^d A(-\rho), \quad \text{that is,} \quad \alpha_j \mapsto (-1)^j \alpha_j. \quad (6.1)$$

Proposition 6.1 shows that ι is the reflection $z \mapsto -z$ written in the coordinates α_j . It reads the wave v_A in the opposite direction and then restores the normalization: it moves the zero limit back to the end $e^z \rightarrow 0$, and for odd p it changes the sign of the profile.

Proposition 6.1 (Reflection). *For every monic A of degree d ,*

$$S_{\iota A}(\rho) = -S_A(-\rho) - 2A(0)A(-\rho), \quad R_{\iota A}(\rho) = -R_A(-\rho). \quad (6.2)$$

Consequently ι maps solutions of the matching equations to solutions, and simple solutions to simple solutions. At a point of $\mathcal{X}_p$,

$$v_{\iota A}(z) = (-1)^p (v_A(-z) - s_p A(0)), \quad P_{\iota A}(\rho) = (-1)^p (P_A(-\rho) + s_p A(0)). \quad (6.3)$$

The pairs A and ιA are the mirror images that N_p counts separately. By (6.3) the mirror image solves the equation in which the terms of one parity have changed sign. For $p = 3$ the involution is $(a, b) \mapsto (-a, b)$ in Table 2. For $p = 2$ it exchanges the two fronts $a = \pm 1$ of (8.2), $12Q(2 - Q)$ and $-12Q^2$: indeed $12(1 - Q)(1 + Q) - 12 = -12Q^2$.

Proof. Since $Q(-z) = 1 - Q(z)$, and D changes sign under $z \mapsto -z$,

$$(A(D)Q)(-z) = A(0) - A(-D)Q = A(0) - (-1)^d (\iota A)(D)Q.$$

This is the first formula in (6.3). Evaluate (3.4) at $-z$, and use (3.4) for ιA on the left:

$$A(0)^2 - 2A(0)A(-D)Q - S_{\iota A}(D)Q = -S_A(0) + S_A(-D)Q.$$

The constant terms cancel because $S_A(0) = -A(0)^2$ (Section 3.4). The remaining terms, and the injectivity of $F \mapsto F(D)Q$, give the first identity in (6.2). Now substitute $S_A = AC_A + R_A$ at $-\rho$:

$$S_{\iota A}(\rho) = (\iota A)(\rho) (-1)^p (C_A(-\rho) + 2A(0)) - R_A(-\rho).$$

Division by the monic polynomial ιA is unique. Hence $R_{\iota A}(\rho) = -R_A(-\rho)$ and $C_{\iota A}(\rho) = (-1)^p (C_A(-\rho) + 2A(0))$. The first relation reads $R_r(\iota\alpha) = (-1)^{r+1} R_r(\alpha)$, so the invertible linear map (6.1) takes each matching equation to plus or minus a matching equation. It therefore maps solutions to solutions, and simple solutions to simple solutions. The second relation and (3.3) give the formula for $P_{\iota A}$. $\square$

A pair is fixed by ι when $\alpha_j = 0$ for every odd j . We write $\mathcal{X}_p^{\text{sym}}$ for the set of these pairs, and put

$$F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}. \quad (6.4)$$

Its first values are 1, 2, 3, 6, 10, 20 for $p = 2, \dots, 7$; Table 1 lists the equations to which they belong. On the subspace $\{\alpha_j = 0, j \text{ odd}\}$ we have $\iota A = A$, and (6.2) shows that R_A is an odd polynomial. The equations $R_r = 0$ with r even therefore hold identically there, and the *matching equations of the subfamily* are the equations $R_r = 0$, r odd, in the unknowns α_j , j even. Their solutions are the pairs of $\mathcal{X}_p^{\text{sym}}$.

Theorem 6.2 (Count in the two pure subfamilies). (i) *For every p , a pair A coincides with its mirror image, $\iota A = A$, if and only if P_A is reflection-symmetric. Thus the pairs that coincide with their mirror image, the pairs of $\mathcal{X}_p^{\text{sym}}$, are exactly the rational-exponential pairs of the purely dispersive equations when p is even, and of the purely dissipative equations when p is odd.*

(ii) *Let $\ell = 2p - 1$ be prime. If A has label L , then ιA has label $-L$. Hence $\iota A = A$ if and only if $L = -L$, and there are exactly F_p such pairs. Each of them is a simple solution of the matching equations of its subfamily.*

(iii) *The rational-exponential pairs of a purely dispersive equation are pulses, for every even p . Those of a purely dissipative equation are fronts: when $2p - 1$ is prime, and for pairs with real coefficients at every odd p .*

In Table 2 the two pairs with $a = 0$ are the $F_3 = 2$ pairs of the purely dissipative subfamily, the Kuramoto–Sivashinsky equation without dispersion. Their labels $\{1, 4\}$ and $\{2, 3\}$ satisfy $L = -L$. Both are fronts; the pulses $(\pm 1, 0)$ need dispersion. For even p see the Kawahara family (Section 8.3).

Remark 6.3 (Every order). At every order the isolated solutions with $\iota A = A$ of the matching equations of the subfamily, counted with multiplicity, number at most F_p (companion note, Theorem 4.1; the weights of the restricted system give the Bézout number F_p). Section 9 records what is known when $2p - 1$ is not prime.

Proof of Theorem 6.2. (i). If $\iota A = A$, then (6.3) gives $P_A(\rho) = (-1)^p (P_A(-\rho) + s_p A(0))$. For even p the polynomial A is odd, so $A(0) = 0$ and P_A is even. For odd p the sum $P_A(\rho) + P_A(-\rho)$ is the constant $-s_p A(0)$. In both cases P_A is reflection-symmetric, and

$$s_p A(0) = -2P_A(0) \quad (p \text{ odd}). \quad (6.5)$$

Conversely, let $P = P_A$ be reflection-symmetric, and apply Corollary 5.3 at the pole $i\pi$, using the period $2\pi i$. If p is even, then $v_A(-z) = v_A(z)$. Letting $z \rightarrow -\infty$ gives $A(0) = 0$, and (6.3) becomes $v_{\iota A} = v_A$. If p is odd, then $-v_A(-z) = v_A(z) + 2P(0)$, and (6.3) becomes $v_{\iota A}(z) = v_A(z) + 2P(0) + s_p A(0)$. Letting $z \rightarrow -\infty$ shows that the constant vanishes. In both cases $v_{\iota A} = v_A$, hence $\iota A = A$ by the uniqueness of the form (3.1).

(ii). The map (6.1) negates the roots of $\bar{A}$, so the label of ιA is $-L$. A label determines its pair (Theorem 3.4), so $\iota A = A$ exactly when $L = -L$. To count these labels, note that the nonzero residues modulo $\ell = 2d + 1$ form d pairs $\{s, -s\}$. A set $L = -L$ of d elements consists of $d/2$ such pairs if d is even, and of 0 and $(d - 1)/2$ pairs if d is odd. In both cases there are $\binom{d}{\lfloor d/2 \rfloor} = F_p$ sets.

Differentiating the relation $R_r(\iota\alpha) = (-1)^{r+1}R_r(\alpha)$ at a fixed point shows that $\partial R_r/\partial\alpha_j = 0$ unless $j \equiv r + 1$ modulo two. The Jacobian matrix of the full system therefore splits into two square blocks. It is invertible, so both blocks are, and one of them is the Jacobian of the equations of the subfamily.

(iii). For even p the polynomial A is odd, so $A(0) = 0$. Let p be odd. If $2p - 1$ is prime, a label with $L = -L$ has an even number d of elements, so it does not contain 0, and the pair is a front by Proposition 3.6. For a general odd p , suppose that a point of $\mathcal{X}_p^{\text{sym}}$ with real coefficients is a pulse, $A(0) = 0$. Then $P_A(0) = 0$ by (6.5). Now v_A is real on the real axis, and it tends to zero with all its derivatives at both ends. Integrating (1.1) along the real axis, and using $P_A(0) = 0$, gives $\int v_A^2 dz = 0$. Hence $v_A = 0$, which is impossible. $\square$

Elliptic pairs of purely dispersive equations. Elliptic waves occur in both subfamilies; the distinction between pulses and fronts does not apply to them. For even p the count of Theorem 4.1 restricts to the subfamily.

Corollary 6.4. *Let p be even, with $2p - 1$ prime. On every lattice, the elliptic pairs whose operator is an even polynomial are those with $u_j = 0$ for odd j in (4.2), and counted with multiplicity there are*

$$2F_p \tag{6.6}$$

of them. On a generic lattice they are distinct.

Proof. An even operator has an even profile, by Corollary 5.3, and the profile (4.2) is even exactly when $u_j = 0$ for odd j . Conversely, if the profile is even, the odd part of (1.1) is $\sum_{j \text{ odd}} a_j D^j V = 0$; the functions $D^j V$ have distinct pole orders, so $a_j = 0$ for odd j . We follow the proof of Theorem 4.1 on the subspace $u_j = 0$, j odd. There the residual of Step 1 is even, and there remain the unknowns and the equations of even weight in the table (4.5) of Step 2: the weights $2, 4, \dots, p - 2, p$ and $p + 2, p + 4, \dots, 2p - 2, 2p$, with $p/2$ of each. The omitted leading forms vanish identically on the subspace, so the remaining ones have only the trivial common zero, by Step 3. Lemma 4.2 therefore applies to the restricted system, and multiplicities refer to this system. For the nodal parameters (4.6), reflection is $Q \mapsto 1 - Q$. An even nodal profile is $\beta + v_A$ with $\iota A = A$, so the computation of Step 5 gives the system (4.7) with $A \in \mathcal{X}_p^{\text{sym}}$: the matching equations of the subfamily, and the quadratic for β . By Theorem 6.2 there are F_p such A , all pulses, each a simple solution of the equations of the subfamily, and $P_A(0) \neq 0$ by Proposition 3.6. For the nodal parameters the restricted system thus has $2F_p$ solutions, all simple, and the rest of Step 5 applies to it without change. $\square$

For odd p there is no fixed-lattice analogue. A purely dissipative equation has a profile that is $-P(0)$ plus a function odd about the pole (Corollary 5.3), and its matching system has one

equation more than it has unknowns. An exact computation on a sample lattice (the script `elliptic_dissipative.py` of the ancillary files, Appendix A) shows that for $p = 3, 5, 7$ it has no solution there, and hence none on a generic lattice: rigidity holds at these orders (Section 9), so Lemma 4.2 applies as in Step 4 of the proof of Theorem 4.1, and the elimination in the proof of part (iii) of the lemma shows that the parameters at which such pairs exist form an algebraic subset of the (g_2, g_3) -plane, invariant under rescaling as in the proof of part (b) of that theorem. These are the two entries 0 of Table 1. At $p = 3$ such pairs occur exactly when $g_2 = 0$: a purely dissipative operator is $P = D^3 + \mu D + a_0$, that is, $B = 0$ in (8.3), which then forces $\mu = 0$ and $g_2 = 0$. At $p = 5$ elliptic waves of the Nikolaevskiy equation on special lattices are given by Kudryashov [44] and Kudryashov and Demina [48].

7 What the count means for a single equation

The count lets the operator vary. We now read it for one evolution equation (2.1) with fixed complex coefficients. The equation is invariant under translations and under Galilean shifts.

Proposition 7.1 (The waves of a fixed equation). *Let $\kappa \neq 0$. Up to a translation and a Galilean shift, the travelling waves of (2.1) that are rational in $e^{\kappa(x-ct)}$, with one pole modulo $(2\pi i/\kappa)\mathbb{Z}$, are*

$$u = b_p \kappa^p v_A(\kappa(x - ct)), \quad c = -b_p \kappa^p P_A(0), \quad (7.1)$$

where A ranges over the points of $\mathcal{X}_p$ that satisfy

$$\frac{b_j}{b_p} = \kappa^{p-j} [\rho^j] P_A, \quad 1 \leq j \leq p-1. \quad (7.2)$$

The data (A, κ) and $(\iota A, -\kappa)$ describe the same wave, read in the two directions (Section 6).

Proof. Such a wave has a finite limit as $e^{\kappa(x-ct)} \rightarrow 0$ (Section 2.3), and a Galilean shift makes this limit zero. Then $w_* = 0$ in (2.3), and (2.4) becomes (7.2) together with the formula for c . After a translation v has the normalization of Section 2, so $v = v_A$ and $P = P_A$ by Proposition 3.1. The last assertion follows from (6.3): the two waves differ by the constant $b_p \kappa^p s_p A(0)$, which is a Galilean shift. $\square$

Given a pair, (7.2) are $p-1$ conditions on the $p-1$ coefficient ratios b_j/b_p and the one unknown κ : a pair imposes $p-2$ conditions on the equation and fixes the rate. A pair with real coefficients gives a real wave of a real equation, smooth on the real axis, exactly when κ is real. Let $\mathbf{a}(A) = ([\rho^{p-1}]P_A, \dots, [\rho^1]P_A)$ be the vector of lower coefficients of P_A , and let $\mathbf{b} = (b_{p-1}/b_p, \dots, b_1/b_p)$ be the coefficient vector of the equation. Condition (7.2) says that $\mathbf{b}$ is obtained from $\mathbf{a}(A)$ by multiplying its k th entry by κ^k , for some $\kappa \neq 0$. As κ varies these vectors trace a curve, denoted $\mathbb{C}^* \mathbf{a}(A)$, and rescaling x moves an equation along it. This is the reason for fixing the rate: without it every pair would be a one-parameter family. Each of the N_p pairs marks one curve, and several pairs may mark the same curve.

Corollary 7.2 (Solvable equations). *Assume that the matching equations have finitely many solutions. This holds when $2p-1$ is prime, and whenever rigidity holds (Proposition 3.2), hence for every $p \leq 573$ (Section 9).*

- (i) Equation (2.1) has a travelling wave that is rational in an exponential, with one pole per period, if and only if $\mathbf{b}$ lies on one of the curves $\mathbb{C}^* \mathbf{a}(A)$, $A \in \mathcal{X}_p$. There are at most $(N_p + F_p)/2$ such curves.

- (ii) If $\mathbf{b} \neq 0$, there are finitely many such waves up to translations and Galilean shifts.
- (iii) If $p \geq 3$, the curves form a set of dimension at most one in $\mathbb{C}^{p-1}$. A generic equation (2.1) has no such wave.

Proof. Part (i) restates Proposition 7.1. For the bound, $\mathbf{a}(\iota A) = (-1) \cdot \mathbf{a}(A)$ by (6.3), so A and its mirror image ιA give the same curve. The number of classes $\{A, \iota A\}$ is half the number of points plus half the number of fixed points. The solutions are finite in number, hence isolated, so these two numbers are at most N_p and F_p (Proposition 3.2, Remark 6.3). For (ii), if $\mathbf{b} \neq 0$ and $\mathbf{b} \in \mathbb{C}^* \mathbf{a}(A)$, some coordinate $[\rho^j]P_A$ is nonzero, and (7.2) leaves at most $p - j$ values of κ . Part (iii) is a dimension count. $\square$

When $\mathbf{b} = 0$ the equation is invariant under the scaling itself, and its waves come in one-parameter families of rates. At order two there are two curves, and both carry waves: $\mathbf{b} = 0$ is the KdV equation, with its solitons of all rates, and $\mathbf{b} \neq 0$ consists of the equations with a nonzero Burgers coefficient, each with one family of fronts, at the rate fixed by that coefficient (Section 8.1).

At $p = 3$, write $P = D^3 + BD^2 + \mu D + a_0$ and use (B, μ) as the coordinates on $\mathbb{C}^2$. Rescaling multiplies B by κ and μ by κ^2 , so the curves are $B^2 = \sigma^2 \mu$ with the origin removed. For the ten pairs $A = \rho^2 + a\rho + b$ of Table 2, (3.11) gives $B = 4a$ and $\mu = a^2 + 19b$, hence the four classical values

$$\sigma^2 = \frac{16a^2}{a^2 + 19b} \in \{0, 16, 144/47, 256/73\}, \quad \text{that is,} \quad |\sigma| \in \{0, 4, 12/\sqrt{47}, 16/\sqrt{73}\}, \quad (7.3)$$

which are those of Kudryashov [32, Eq. (39)]. In the normalization of the Introduction, with $b_1 = b_3 = 1$ and $b_2 = \sigma$, condition (7.2) reads $\kappa^2 = 1/(a^2 + 19b)$ and $\sigma = 4a\kappa$. Here the bound $(N_3 + F_3)/2 = 6$ is not attained, because the two pairs with $a = 0$ share $\sigma = 0$, and $(\pm 1, 0)$ and $(\pm i, 0)$ share $\sigma^2 = 16$. For coefficients off these curves, approximate solutions close to the exact ones are constructed by Kudryashov and Kochanov [57].

For odd p the classification turns Corollary 7.2 into a statement about all meromorphic travelling waves.

Corollary 7.3 (Generic equations with p odd). *Let $p \geq 5$ be odd and assume rigidity (R_{p-1}), which holds when $2p - 1$ is prime and for every $p \leq 573$. The coefficient vectors $\mathbf{b} \in \mathbb{C}^{p-1}$ for which (2.1) has a nonconstant meromorphic travelling wave lie in an algebraic subset of dimension at most two. In particular a generic equation has none.*

Proof. By Theorem 5.1, such a wave is rational, rational in an exponential, or elliptic, with one pole per period. Take $\kappa = 1$ in (2.3), so that $[\rho^j]P = b_j/b_p$ for $1 \leq j \leq p - 1$. The rational-exponential waves give a set of dimension at most one, by Corollary 7.2. A rational wave is, after a translation, of the form (2.8). By Lemma 3.3 it comes from a monic A of degree $p - 1$ that divides H_A . By rigidity, $A = \rho^{p-1}$, hence $P = D^p$ and $\mathbf{b} = 0$. For the elliptic waves, Steps 1–3 of the proof of Theorem 4.1 use only rigidity, and with them Lemma 4.2(i) shows that every system $\mathcal{E}_p(g_2, g_3)$ has finitely many solutions. The solutions for all (g_2, g_3) together therefore form an algebraic set of dimension at most two, and so does the closure of its image under $P \mapsto \mathbf{b}$. $\square$

At $p = 3$ the argument gives the exact answer: the set of these $\mathbf{b} = (B, \mu)$ is the union of the four curves $B^2 = \sigma^2 \mu$ of (7.3). Indeed the elliptic waves require $B^2 = 16\mu$ (Section 8.2), which is one of the four curves, and the point $\mathbf{b} = 0$ of the rational wave lies on all of them.

The same dimension count applies to the one-pole waves of every order $p \geq 3$ for which rigidity holds: a generic equation has no travelling wave that is rational, rational in an exponential, or elliptic with one pole per period. For even p , however, Theorem 5.1 excludes other meromorphic waves only when the equation is purely dispersive.

Can one equation have two different waves of this kind at the same rate and speed? It cannot, and this holds for every P , including the mixed equations of odd order, to which Theorem 5.1 does not apply: the argument only uses the leading coefficient of a convolution.

Proposition 7.4 (One profile per operator). *Let A_1, A_2 be monic polynomials of degree $p - 1$ satisfying $A_1 \mid S_{A_1}$ and $A_2 \mid S_{A_2}$. If $P_{A_1} = P_{A_2}$, then $A_1 = A_2$. Consequently a monic operator of degree p admits at most one normalized rational-exponential one-pole profile. Any count of distinct pairs with this normalization is therefore also a count of distinct operators.*

Proof. Suppose $P_{A_1} = P_{A_2} = P$, and put $h = A_1 - A_2$. If $h \neq 0$, its degree n is less than $p - 1$; write $f \neq 0$ for its leading coefficient. By (3.3) and the bilinearity and symmetry of the convolution, $S_{A_1} - S_{A_2} = h \star (A_1 + A_2) = (2/s_p)Ph$. The power-sum calculation in (3.2) gives the leading coefficient $2f \int_0^1 t^n (1 - t)^{p-1} dt$ on the left. Since P is monic, the right side has leading coefficient $(2/s_p)f$, and $2/s_p$ is the beta integral of (3.2). Cancelling f would therefore give

$$2 \int_0^1 t^n (1 - t)^{p-1} dt = \int_0^1 t^{p-1} (1 - t)^{p-1} dt.$$

This is impossible: $n < p - 1$ implies $t^n > t^{p-1}$ for $0 < t < 1$, so the left side exceeds twice the right side. Thus $h = 0$. $\square$

Every P_A has a root among $1, \dots, p$. Let m be the least positive integer with $A(m) \neq 0$; then $m \leq p$, because A has degree $p - 1$. As $e^z \rightarrow 0$ the profile begins with $s_p(-1)^{m-1}A(m)e^{mz}$ and its square with e^{2mz} , so the coefficient of e^{mz} in (1.1) gives $P_A(m) = 0$. In physical terms, the wave leaves its background like e^{mz} , so m is a spatial eigenvalue of the linearization about the background, that is, a root of P ; and m is one of $1, \dots, p$.

Other equations with the same profile equation. The reduction of Section 2 uses only the travelling-wave variable. The counts therefore apply to any equation $\mathcal{L}u + \mathcal{M}(u^2) = 0$ with constant-coefficient operators, mixed derivatives and several space variables included, whose plane-wave reduction integrates to (1.1); the coefficients of P then depend on the speed and the direction of the wave. An example in three space variables is the equation of Kudryashov and Sinelshchikov [59] for waves in a liquid with gas bubbles, whose travelling-wave reduction is that of KdV–Burgers. The repository lists such equations by profile operator, with their reductions (directory `tools/wave_dictionary/`).

Ansatz methods. Many exact travelling waves in the literature are produced by substituting a finite expansion: the tanh and sech methods, the exp-function and (G'/G) -expansion methods, the simplest-equation method with a Riccati equation [45], and the Jacobi and Weierstrass elliptic-function expansions [36]. In their standard form, with one phase $kx - \omega t$ and constant coefficients, the first group produces functions that are rational in one exponential and the second group elliptic functions of one argument; rational functions of z arise as limits. With respect to its primitive period, an output with one pole per period is, up to the symmetries of Proposition 7.1, one of the waves (7.1), one of the elliptic waves of Section 4, or the rational wave; for the equations of Theorem 5.1 every nonconstant output has one pole per period. The counts therefore serve

as a checklist: at a covered order, once the rate or the period lattice is fixed, a tanh-type or elliptic ansatz with one pole per period can return at most N_p , respectively M_p , normalized solutions, independently of the expansion method. This agrees with the critical comparisons of these methods [36, 41, 50]: the “new” waves they returned for the KdV–Burgers and Kuramoto–Sivashinsky equations coincide with the known ones [41, 46], and an ansatz whose pole order is lower than p returns no wave at all [47]. For the conformable derivative $T_\alpha f(t) = t^{1-\alpha} f'(t)$, $t > 0$, $0 < \alpha \leq 1$ [29], the change $\tau = t^\alpha/\alpha$ gives $T_\alpha = d/d\tau$, and similarly for a spatial conformable derivative, so whenever the transformed equation has the profile equation (1.1) its waves fall under the same counts; this does not apply to the Caputo or Riemann–Liouville derivatives.

8 Examples

The examples show the counts of Table 1 at work and connect them with the classical lists. Order three was begun in Section 3.2; the normalizations are those of Section 2.

8.1 Order two: KdV–Burgers

The integrated travelling-wave equation for $u_t + uu_x + \nu u_{xx} + \delta u_{xxx} = 0$, $\delta \neq 0$, has order $p = 2$; the coefficient ν may have either sign. Its exact front solutions have been studied in several equivalent forms [26, 31, 36, 46, 72]. The equation arises, for instance, for waves in a liquid with gas bubbles [52] and in a viscoelastic tube [43]. For $A(\rho) = \rho + a$ the matching system is the single equation of the Introduction. Polynomial division gives

$$\text{rem}_A S_A = -\frac{a(a-1)(a+1)}{6}, \quad P_A = \rho^2 + 5a\rho + a^2 - 6a - 1, \quad v_A = 12((a+1)Q - Q^2). \quad (8.1)$$

Thus there are exactly $N_2 = 3$ simple pairs, as Theorem 3.4 states for $2p - 1 = 3$:

a	$P_A(\rho)$	v_A
-1	$(\rho - 2)(\rho - 3)$	$-12Q^2$
0	$\rho^2 - 1$	$12Q(1 - Q) = 3 \operatorname{sech}^2(z/2)$
1	$(\rho - 1)(\rho + 6)$	$12Q(2 - Q)$.

(8.2)

The pulse $a = 0$ is the KdV soliton, the single pair of the purely dispersive subfamily, $F_2 = 1$. The two fronts belong to mixed equations, and the reflection ι exchanges them: for a given equation with $\nu \neq 0$ they are one and the same front, read in the two directions (Proposition 7.1), and $N_2 = 3$ counts both readings. In the original variables, (2.3) and (2.4) read

$$u(x, t) = w_* + \delta \kappa^2 v_A(\kappa(x - ct)), \quad \nu = \delta \kappa(5a), \quad w_* - c = \delta \kappa^2(a^2 - 6a - 1).$$

In particular, a nonzero ν fixes the front rate to $\kappa = \nu/(5a\delta)$, $a = \pm 1$, in agreement with the classical parameter restriction. The elliptic pairs at this order are (4.4): the two backgrounds of one cnoidal wave, both for the KdV equation. The classification of Theorem 5.1 applies to the KdV subfamily $a_1 = 0$; for the mixed operators the same conclusion cannot be expected (Remark 5.2).

8.2 Order three: elliptic pairs, and the lattices on which they merge

An elliptic solution must have zero residue at its pole. For $P = D^3 + BD^2 + \mu D + a_0$ the Laurent residue is $60\mu/19 - 15B^2/76$, so $B^2 = 16\mu$; this is the curve $\sigma^2 = 16$ of (7.3). With this restriction

the entire elliptic matching scheme is

$$\begin{aligned} P &= D^3 + BD^2 + \frac{B^2}{16}D + a_0, & v &= -60\wp' - 15B\wp - \frac{B^3}{64} - a_0, \\ B^4 &= 3072g_2, & a_0^2 &= \frac{13B^6}{2048} - 2160g_3. \end{aligned} \tag{8.3}$$

Read with the operator fixed, this is the classical solution of Kudryashov [33], see also Kudryashov [34]: $w = v + a_0$ solves the usual integrated equation $w''' + Bw'' + \mu w' + w^2/2 + K = 0$ with $K = -a_0^2/2$. For a given equation with $B^2 = 16\mu$, the value of g_2 is fixed by B and g_3 is free, so the equation has a one-parameter family of elliptic waves. Necessity of the restriction $B^2 = 16\mu$ is proved by Hone [24], and the classification of all meromorphic KS solutions is completed by Eremenko [22]. For the count we read (8.3) the other way: fix g_2, g_3 and solve for B and a_0 .

On a generic lattice, the equation for B has four solutions, and the equation for a_0 then gives two choices for each of them: the two backgrounds $\pm a_0$ of (2.5). These are the $M_3 = 8$ pairs. When the lattice changes, some of the pairs can meet. The multiplicity of the resulting point records how many nearby pairs have come together, so the count with multiplicity can remain eight even when fewer than eight distinct pairs remain. Pairs meet exactly where the Jacobian of the last two equations of (8.3) vanishes, and that Jacobian is a nonzero constant times $B^3 a_0$. In terms of the invariant $j(\Lambda)$ of Section 4: if $B = 0$, then $g_2 = 0$, hence $j = 0$; if $a_0 = 0$ on a lattice, elimination of B gives $675g_3^2 = 169g_2^3$, that is, $j = -300$. These conditions are sufficient as well. Thus:

$j(\Lambda)$	distinct pairs	total with multiplicity
$j \notin \{0, -300\}$	8	8×1
-300	6	$2 \times 2 + 4 \times 1$
0	2	2×4 .

(8.4)

The two exceptional cases show the two ways in which pairs can meet. At $j = -300$, the four values of B remain distinct. For two of them, the two choices of a_0 meet at zero: the two backgrounds of one wave coincide, in one pair of multiplicity two. For instance, $g_2 = 1/12$, $g_3 = 13/1080$ gives the double points $(B, a_0) = (\pm 4, 0)$. At $j = 0$, all four values of B meet at $B = 0$. The two roots of $a_0^2 = -2160g_3$ remain distinct, since $g_3 \neq 0$, so there are two pairs, each of multiplicity four. Both have the operator $D^3 + a_0$, which is purely dissipative.

The count stays eight for an exact reason. The unknowns eliminated on the way to (8.3) are determined linearly, with nonzero rational pivots, so its last two equations carry the full matching system, multiplicities included; the first has four roots B and, for each of them, the second has two roots a_0 , counted with multiplicity, which gives eight solutions on every lattice. This is the form in which Theorem 4.1 holds on an arbitrary lattice, and (8.4) is the exceptional set of that theorem at order three.

8.3 Purely dispersive equations: Kawahara and seventh-order KdV

Order four. For the Kawahara equation [18, 27], the integrated, shifted and monic travelling-wave operator is

$$P = D^4 + a_2 D^2 + a_0. \tag{8.5}$$

The classification for this equation, using its first integral (5.5), is already established by Demina and Kudryashov [18]; see also Ye et al. [73]. By Theorem 6.2, the pairs with an even operator

are those with $\iota A = A$, that is, $A = \rho^3 + \alpha_2 \rho$. All of them are pulses: the profile is even about its pole, $v(Q) = v(1 - Q)$. The quotient and remainder are

$$\begin{aligned} P_A &= \rho^4 + 13\alpha_2 \rho^2 + \frac{31\alpha_2^2 - 70\alpha_2 + 7}{3}, \\ v_A &= 280(1 + \alpha_2)Q(1 - Q) - 1680Q^2(1 - Q)^2, \\ \text{rem}_A S_A &= -\frac{\rho}{420}(\alpha_2 + 1)(31\alpha_2^2 - 31\alpha_2 + 10). \end{aligned} \quad (8.6)$$

The three distinct roots, $\alpha_2 = -1$ and $\alpha_2 = 1/2 \pm 3i/(2\sqrt{31})$, give $F_4 = \binom{3}{1} = 3$ simple pairs. The cubic already appears in Kudryashov [32, Eq. (47)], with $y = -1/\alpha_2$, where one real and two complex conjugate roots are noted. The three values are the simply periodic parameter choices of Demina and Kudryashov [18, Section 3], in the present normalization: the parameter in their Eq. (31) equals $1/\alpha_2$ here. At $\alpha_2 = -1$,

$$v = -105 \operatorname{sech}^4(z/2), \quad P = (D^2 - 4)(D^2 - 9). \quad (8.7)$$

This is the familiar solitary wave of the Kawahara equation; the other two pairs are complex at unit real rate. The unrestricted order-four count is $N_4 = 35$. Of these pairs, 21 have real coefficients, and all 21 have rational coefficients: five pulses and sixteen fronts, which are eleven wave shapes up to reflection. They are the one-pole waves of the mixed fifth-order equations, such as the Benney–Lin equation and the equation of Kudryashov and Sinelshchikov [53] for a viscoelastic tube, and the repository dictionary lists them; solitary waves of this family are found in Kudryashov and Demina [49].

All elliptic members of (8.5) have the form (8.8) below, and substitution gives exactly the two conditions (8.9) and (8.10):

$$v = -1680g_2^2 - \frac{280a_2}{13}g_2 + 168g_2 + \frac{31a_2^2}{507} - a_0, \quad (8.8)$$

$$0 = 31a_2^3 - 42588a_2g_2 - 4745520g_3, \quad (8.9)$$

$$a_0^2 = 23184g_2^2 + \frac{7980}{169}a_2^2g_2 + \frac{101520}{13}a_2g_3. \quad (8.10)$$

These are the Weierstrass profiles and parameter relations obtained from Laurent series by Demina and Kudryashov [18, Section 3], expressed in the normalization (1.1): in their notation $a_2 = -\beta$, $a_0^2 = C_0^2 + 12C_1$ and $v = -6(w - w_*)$, where their w_* is a constant equilibrium and $a_0 = C_0 - 6w_*$. The first condition is cubic in a_2 , and the second is monic quadratic in a_0 , so there are $3 \times 2 = 6$ solutions, counted with multiplicity, on every fixed lattice: three values of a_2 , each with its two backgrounds. This is the number $2F_4$ of Corollary 6.4. At $(g_2, g_3) = (1, 0)$, the cubic has three distinct roots and the right side of (8.10) is nonzero at all of them, so the six pairs are simple on a nonempty open set of lattices. The unrestricted elliptic count at this order is $M_4 = 30$; elliptic formulas with nonzero odd derivative coefficients in the full fifth-order evolution equation are given by Kudryashov and Sinelshchikov [58].

Order six. The seventh-order KdV equation has the operator $P = D^6 + a_4D^4 + a_2D^2 + a_0$; its meromorphic travelling waves are studied by Dang [12]. Here $A = \rho^5 + \alpha_2\rho^3 + \alpha_4\rho$ and $s_6 = 5544$. The remainder of S_A has two coefficients, at ρ^3 and at ρ , which are polynomials in α_2, α_4 of weights eight and ten; the ancillary script prints them. Eliminating α_4 , which is determined by α_2 at every solution, gives

$$(\alpha_2 + 5)(\alpha_2 + 10)(49\alpha_2^2 - 5\alpha_2 + 10) \varphi_6(\alpha_2) = 0, \quad (8.11)$$

where φ_6 is an irreducible sextic without real roots, also printed by the script. The ten roots are distinct, in agreement with $F_6 = \binom{5}{2} = 10$. Two pairs are real:

$$\begin{array}{c|c|c} A & P_A & v_A \\ \hline \rho(\rho^2 - 1)(\rho^2 - 4) & (\rho^2 - 9)(\rho^2 - 16)(\rho^2 - 25) & 10395 \operatorname{sech}^6(z/2) \\ \rho(\rho^2 - 1)(\rho^2 - 9) & (\rho^2 - 4)(\rho^2 - 25)(\rho^2 - 71) & 10395 \operatorname{sech}^4(z/2)(1 + \operatorname{sech}^2(z/2)). \end{array} \quad (8.12)$$

All ten pairs are pulses. The two real pairs are the solitary waves of Ryabov et al. [70]. An exact computation on a sample lattice gives $2F_6 = 20$ for the elliptic pairs of this family.

8.4 Purely dissipative equations of order five

At $p = 5$ the purely dissipative operators are $P = D^5 + a_3 D^3 + a_1 D + a_0$, and $A = \rho^4 + \alpha_2 \rho^2 + \alpha_4$. All 126 pairs at $p = 5$ are simple (Section 9), so there are exactly $F_5 = 6$ distinct pairs here. They are the three pairs in the first line of

$$\begin{aligned} A = (\rho^2 - 1)(\rho^2 - 4), \quad A = (\rho^2 - 1)(\rho^2 - 9), \quad A = (\rho^2 - 1)\left(\rho^2 + \frac{1}{5}\right), \\ 1105\alpha_2^3 - 2969\alpha_2^2 + 5570\alpha_2 - 4000 = 0, \quad \alpha_4 = \frac{5\alpha_2^3 + 21\alpha_2 - 10}{79\alpha_2 + 210}, \end{aligned} \quad (8.13)$$

and the three pairs given by the second line. The cubic has one real root. All six pairs are fronts: $A(0) = \alpha_4 \neq 0$ at each of them. For the four real ones this is Theorem 6.2(iii); for all six it follows from Proposition 9.1(iii), since all 126 pairs at $p = 5$ are simple (Appendix A). The six pairs are the six solitary waves of Kudryashov and Demina [48]; in their notation $w = 1/(4a_3)$. The four with real coefficients are, after the scales are restored, the solutions (1.4) of Kudryashov and Migita [40]; see also Ryabov et al. [70]. For instance, $A = (\rho^2 - 1)(\rho^2 - 4)$ gives the operator $P_A = \rho^5 - 55\rho^3 + 1254\rho - 2520$ and the profile $v_A = 5040 Q^3(10 - 15Q + 6Q^2)$, a front from 0 to $5040 = -2P_A(0)$.

9 Other orders and open questions

The proofs of Sections 3 and 4 use the prime $\ell = 2p - 1$ twice: to show that solutions cannot escape to infinity, which is rigidity, and to show that every pair is simple. This section records what is known when $2p - 1$ is not prime. The proofs use the weighted form of Bézout's theorem (Proposition 3.2) instead of the labels. They are given in the companion note `anc/general_orders.pdf`, which is part of the source package. The note contains the proof of Proposition 3.2, the general-order versions of the results of Sections 3.4, 4 and 6, and the proof of rigidity when $2p - 1$ is a prime power. It also contains the general lemma on reduction modulo ℓ behind Step 3 of the proof of Theorem 3.4, and a second, algebraic proof of parts (i) and (ii) of Lemma 4.2.

Proposition 9.1 (Every order). *Let $p \geq 2$.*

- (i) *The isolated solutions of the matching equations, counted with multiplicity, number at most N_p (Proposition 3.2), and those with $\iota A = A$, as solutions of the matching equations of the pure subfamily, at most F_p (Remark 6.3).*
- (ii) *If rigidity (R_{p-1}) holds, all the systems below have finitely many solutions, and counted with multiplicity there are exactly N_p rational-exponential pairs (Proposition 3.2), S_p solutions of the pulse equations (3.18), F_p solutions of the matching equations of the pure subfamily, M_p elliptic pairs on every lattice and, for even p , $2F_p$ elliptic pairs with an even operator.*

(iii) *If, in addition, every rational-exponential pair is simple, then all the conclusions of Theorem 3.4(a), Proposition 3.6, Theorem 6.2, Theorem 4.1 and Corollary 6.4 hold at this p , except for the labelling.*

Two hypotheses thus remain when $2p - 1$ is not prime, first at $p = 5$ (Section 8.4). This is what is known about them.

- *Rigidity.* Rigidity (R_{p-1}) is proved when $2p - 1$ is a prime or a prime power. Exact integer certificates in the source package establish it for every $p \leq 73$, that is, for every degree $d = p - 1 \leq 72$, and finite-field Gröbner computations, checked by independent pipelines, for every $p \leq 573$ (repository, directory `certificates/rigidity_572/`). Hence the counts with multiplicity of Proposition 9.1(ii) hold for every $2 \leq p \leq 573$.
- *Simplicity.* At the composite indices below 25, exact computations show that all pairs are simple: at $p = 5$ by a Gröbner basis over $\mathbb{Q}$, at $p = 8$ over a finite field with Hensel lifting, and at $p = 11$ by certificates over the 19-adic numbers (Appendix A). Inside the two pure subfamilies the computation is much smaller, and it shows that the F_p pairs are distinct for every $p \leq 14$.

Corollary 9.2 (Orders up to twelve). *For every $2 \leq p \leq 12$ there are exactly N_p rational-exponential pairs, all simple, of which S_p are pulses and F_p belong to the pure subfamily of that order; and a generic lattice carries exactly M_p distinct elliptic pairs. In particular this holds for all the equations of Table 1.*

At $p = 5, 8$ and 11 this is Proposition 9.1(iii) with the two items above; the other orders up to twelve have a prime index. The first order at which distinctness is not known is $p = 13$.

Open questions. The first question is rigidity (3.8) in every degree; Lemma 3.3 says what it means for rational waves. It is proved in the cases listed above, which include infinite arithmetic families of higher orders. By Proposition 9.1, it would establish both enumeration formulas, as counts with multiplicity, at every order.

The second question is whether every point of $\mathcal{X}_p$ is simple. The elliptic simplicity assertion is generic in the lattice and is derived from this one in Theorem 4.1. At a solution $S_A = AC_A$, the tangent map of the matching equations is

$$J_A(h) = \text{rem}_A(2(A \star h) - C_A h), \quad \deg h < d = p - 1. \quad (9.1)$$

The question is whether this map always has zero kernel. The prime-index argument proves this after reduction modulo ℓ , where the map becomes multiplication by a polynomial coprime to $\overline{A}$; a uniform replacement is missing at composite indices. One might expect simplicity to follow from the uniqueness of the Laurent expansion in Theorem 5.1. It does not: Laurent uniqueness concerns a fixed operator, whereas (9.1) allows the operator to vary with the profile.

Conjecture 9.3. Rigidity (3.8) holds in every degree, and the matching Jacobian (9.1) is invertible at every point of $\mathcal{X}_p$. Consequently there are exactly $\binom{2p-1}{p-1}$ rational-exponential pairs, all distinct, for every $p \geq 2$.

The conjecture would also imply M_p simple elliptic pairs on a generic fixed lattice at every order, F_p distinct pairs in the two pure subfamilies, and that the rational-exponential waves of every purely dissipative equation are fronts (Theorem 6.2). Simplicity on every lattice is false already at order two (4.4); (8.4) describes the full exceptional set at order three. Whether a mixed equation of even order $p \geq 4$ can have meromorphic waves of none of the three kinds is a separate open question (Remark 5.2).

A Exact verification and ancillary material

This appendix describes the computer checks that accompany the paper. The scripts use exact rational or finite-field arithmetic; instructions, dependency versions and checksums are included with the source package. The prime-index theorem does not depend on the computer calculations. The ancillary material contains five kinds of checks.

1. `verify_manuscript.py` reconstructs the discrete and continuous convolutions, the low-order Laurent coefficients, and the Weierstrass differential algebra. It checks the leading-term comparison used in Proposition 7.4, and the displayed order-two, order-three and order-four formulas by exact substitution, including the correspondence of Table 2 with the tables of Conte and Musette [6], Kudryashov and Zargaryan [42], the multiplicities (8.4), the Kawahara coefficient restriction and its literature normalization, the nodal correspondence, and the Fisher identity. These finite substitutions check the displayed formulas; the uniform proofs are the arguments in the text.
2. `p5_count.sing` is a self-contained SINGULAR calculation over $\mathbb{Q}$ for the full matching equations at $p = 5$. Its quotient dimension, which is the count with multiplicity, is 126. Adjoining the Jacobian determinant gives the unit ideal: the determinant vanishes at no solution, so all 126 solutions are simple.
3. At $p = 8$ the same two outputs, 6435 and the unit ideal, are obtained over the finite field $\mathbb{F}_{32003}$. Every solution modulo 32003 is therefore simple and lifts uniquely by Hensel's lemma, as in Step 4 of the proof of Theorem 3.4, and the lifts are all the solutions because the count with multiplicity is 6435, by Proposition 9.1(ii). At $p = 11$ each of the 352716 solutions is certified over the 19-adic numbers, by Hensel's lemma at the simple points of the reduction modulo 19 and by exact Newton-polygon and Newton-Kantorovich certificates at the singular ones. The archived certificate and the record of its validation are in the repository, directory `certificates/`.
4. `rigidity/` supplies the proof and exact certificate for rigidity (R_d) , $d = p - 1 \leq 72$. For $2d + 1 = \ell^e + g$, $g \in \{0, 2, 4, 6\}$, the proof reduces the normalized continuous convolution to a residual system of degree at most g ; the case $g = 0$, a prime power, is also proved in the companion note (Section 9). The certificate retains every residual degree, with integer ideal-membership identities and certified prime factors; the replay reconstructs the inputs, multiplies out every identity and checks all 72 arithmetic choices. It does not infer characteristic-zero nonexistence from a list of residue-field points.
5. `symmetry_checks/` contains the exact checks for the two pure subfamilies and for single equations: the identities of Proposition 6.1 with symbolic coefficients for $d \leq 5$; the labels of Table 2 and of the 21 real pairs at $p = 4$, with the statements of Proposition 3.6 and Theorem 6.2(ii) about them; the counts with multiplicity F_p for $p \leq 8$, and finite-field certificates, of the same kind as at $p = 8$, that the F_p pairs are distinct for every $p \leq 14$; on a sample lattice, the counts with multiplicity $2F_p$ of Corollary 6.4 for $p \leq 6$, and the absence of elliptic pairs of purely dissipative equations for $p = 3, 5, 7$ (`elliptic_dissipative.py`, Section 6); the six pairs (8.13) and the ten pairs (8.11), with the sextic factor in full; and the scaling (7.1) by substitution in the evolution equation.

A Lean 4 formalization accompanies the paper at <https://github.com/migita/binomial-count-meromorphic-waves>, directory `lean/`, with Lean and Mathlib pinned to version 4.33.1. It proves rigidity (R_d) for $2d + 1$ prime (`rigidity_prime_index`) and the count and simplicity statements of Theorem 3.4, both for the matching equations (3.5) (`primeIndexCountAndSimplicity`)

and for the normalized pairs (P, v) as solutions of (1.1) in the field of rational functions of e^z (`primeIndexRationalExponentialCountAndSimplicity`), together with the correspondence between the two formulations for every $p \geq 2$. The discrete convolution is defined there by the Bernoulli-polynomial power sums; the remainder equations and their Jacobian are those of Section 3, and the proof route is that of Remark 3.5. The files `EnumerationCheck.lean` and `Check.lean` verify that the proofs have exactly the stated propositions as their types and that the theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; the project contains no `sorry`, added axiom or `native_decide`.

Tool and computational resource disclosure

This paper was written with substantial use of generative AI, which is disclosed here in the form recommended by the Leiden Declaration on Artificial Intelligence and Mathematics¹ and required by arXiv. No automated system is an author; credit and responsibility for the results, and for the attribution of prior work, rest with the author.

- **gpt-6-astra** (OpenAI), August–September 2026. The main mathematical ideas of the paper, the proofs of Theorems 5.1, 3.4 and 4.1 and of the weighted count in the companion note, the rigidity condition (R_d) and its certificate, the exact verification scripts, the SINGULAR input, the first complete draft of the text, and the Lean 4 formalization of Remark 3.5 were produced by this model in dialogue with the author.
- OpenAI Codex v0.153.4, September 2026. Independent mathematical audit of the final revision, integration of the edits, and rebuilding and validation of the source package.
- Claude Code (Anthropic, Claude Fable 5.1), September 2026. The literature search behind the attributions in Section 1, the readability revision of all sections, and the leading-coefficient argument of Proposition 7.4, which Codex then checked independently. In the final revision it proposed and proved the results on the two pure subfamilies and on single equations (Corollary 5.3, Section 6, Corollary 6.4 and Section 7), with their exact checks, and reorganized the manuscript around them. In the last revision it made the prime-index theorem the spine of the paper: it wrote the proof of Theorem 3.4 in the form used by the Lean formalization, derived Proposition 3.6 and Theorem 6.2 from the labels, replaced the Bézout count in the proof of Theorem 4.1 by a flat-family argument, and moved the general-order material to the companion note; Codex reviewed these arguments. It also audited the Lean formalization independently: the formal statements against the manuscript, a rebuild from source, and the axiom audit. In the present revision it reorganized the exposition for readers in nonlinear science: the counts now precede the classification, the proof of Theorem 3.4 is presented through a dominant-balance argument and Hensel’s lemma, and the proof of Theorem 4.1 through a conservation-of-number lemma, the flat-family argument having moved to the companion note.

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¹<https://leidendeclaration.ai/>, 2 June 2026.

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